Gene Expression Pattern Matching

Evaluating Similarity Metrics for NMF Constraint Patterns

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April 2026

Motivation

- **Goal:** Identify genes whose expression over time matches biologically observed constraint patterns
- **Data:** $\sim$17,856 genes across 4 timepoints (0, 3, 6, 9) in 4 clusters, raw and log-transformed
- **Problem with NMF:** All factors are non-negative, so NMF ranks genes by coefficient magnitude, not shape match. Up-regulation and down-regulation are conflated.
- **Our approach:** Compare 6 similarity metrics that directly measure gene–pattern shape similarity

Constraint Patterns

Cluster	**t=0**	**t=3**	**t=6**	**t=9**	**Shape**
One	1.45	12.0	5.21	6.51	Spike at t=3
Two	0.371	0.803	1.22	1.78	Steady increase
Three	0.041	0.0	0.037	0.035	Near-flat, dip at t=3
Four	0.009	0.638	0.018	0.024	70× spike at t=3

All data min-max normalized to $(0, 1]$ before comparison. Both raw and log-transformed (LT) versions analyzed.

The 6 Metrics at a Glance

Metric	Measures	Ranking	Key property
Pearson	Shape (mean-centered)	Higher = better	Scale & shift invariant
Cosine	Vector angle (raw)	Higher = better	No mean-centering
DTW	Time-aligned distance	Lower = better	Allows temporal warping
Fréchet	Max curve deviation	Lower = better	Worst-point sensitive
MSE	Avg squared error	Lower = better	Simplest distance
sMAPE	Symmetric % error	Lower = better	Relative scale

Pearson Correlation — Primary Metric

$$r = \frac{\sum(p_i - \bar{p})(g_i - \bar{g})}{\sqrt{\sum(p_i - \bar{p})^2 \cdot \sum(g_i - \bar{g})^2}}$$

- Mean-centers both vectors $\Rightarrow$ measures **shape only**
- Scale and shift invariant: $[1, 12, 5, 7]$ and $[100, 1200, 500, 700]$ get the same score
- Best fit for the biological question: "does this gene go up when the pattern goes up?"
- **Caveat:** with $n = 4$, there is a $\sim$20% chance of $|r| > 0.8$ under the null — interpret rankings, not absolute values

Why Cosine Similarity Diverges from Pearson

$$\cos(\theta) = \frac{\sum p_i g_i}{\sqrt{\sum p_i^2}\sqrt{\sum g_i^2}}$$

- Same formula as Pearson but **without mean-centering**
- A flat gene $[1, 1, 1, 1]$ gets high cosine with almost any positive pattern
- Finds housekeeping genes (high, stable expression) instead of shape-matching genes
- Behavior is **context-dependent**: perfect overlap with Pearson on clusterThree (20/20), but only 2/20 on clusterOne

DTW, Fréchet, MSE — Distance Metrics

DTW

Finds optimal temporal alignment. At 4 timepoints, warping is minimal $\Rightarrow$ largely redundant with pointwise methods.

Fréchet

Worst-case pointwise deviation ("dog-walking distance"). At $n = 4$, behaves very similarly to MSE.

MSE

Average squared error. After normalization, approximately inversely related to Pearson: MSE $\approx 2\sigma^2(1 - r)$.

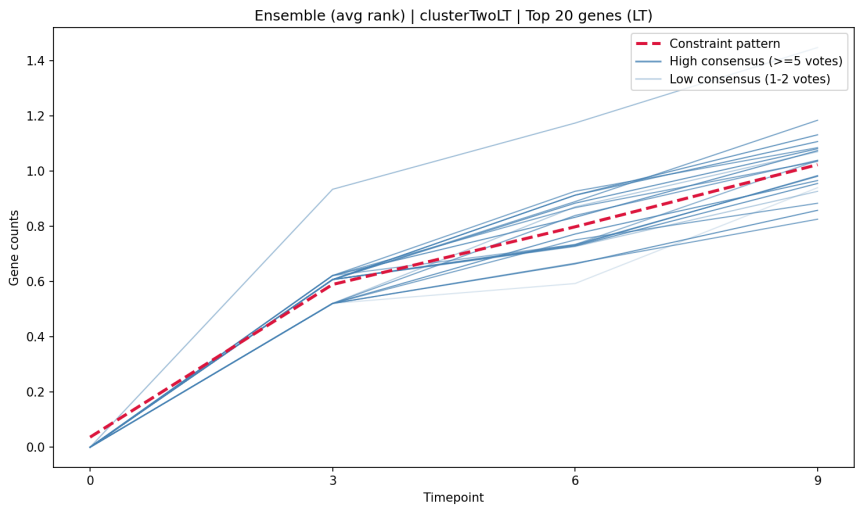
Key insight: Fréchet and MSE are near-redundant (19–20/20 overlap). Both confirm Pearson's findings from a distance perspective.

sMAPE — Failure Mode at Zero Values

$$\text{sMAPE} = \frac{1}{n} \sum \frac{2|p_i - g_i|}{|p_i| + |g_i|}$$

- Measures **relative** (percentage) error per timepoint
- When $p_i = 0$: $\text{sMAPE}_i = 2|g_i|/|g_i| = 2.0$ regardless of g_i
- clusterThree has a zero at $t = 3 \Rightarrow$ sMAPE cannot discriminate at the most informative timepoint
- Result: **0/20 overlap** with every other method on clusterThree

Results: clusterTwo LT — Best Agreement



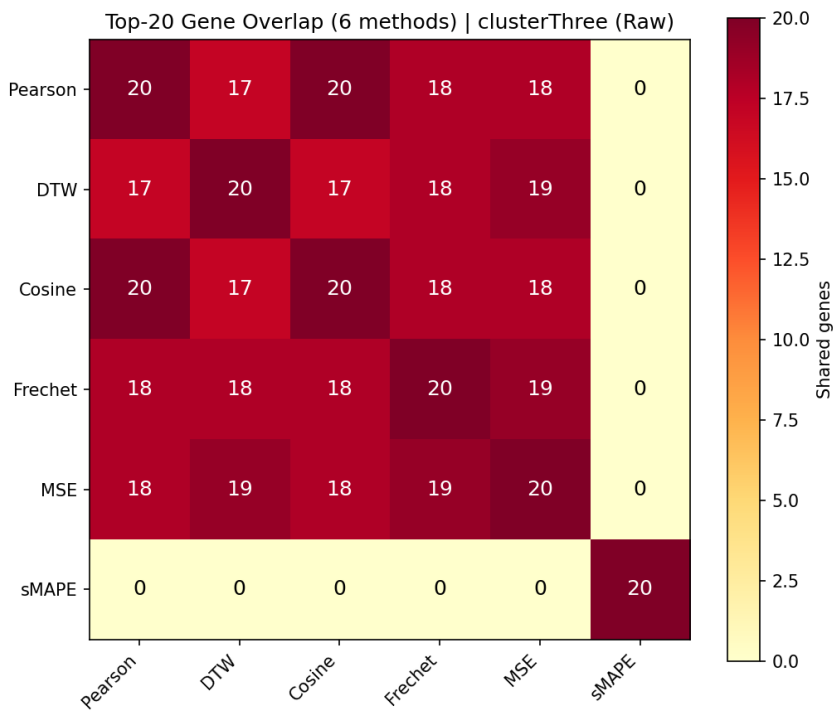
9 of top 11 genes achieve unanimous 6/6 votes:

RAB14, SYNGR1, ARHGAP25,
HNRNPH3, BSG, CAPN2,
NOL7, ZC3H15, PRDX6

Steady-increase pattern is easiest to match — all methods converge.

Highest-confidence candidates
for biological follow-up.

Results: clusterThree — Strong Consensus (Except sMAPE)



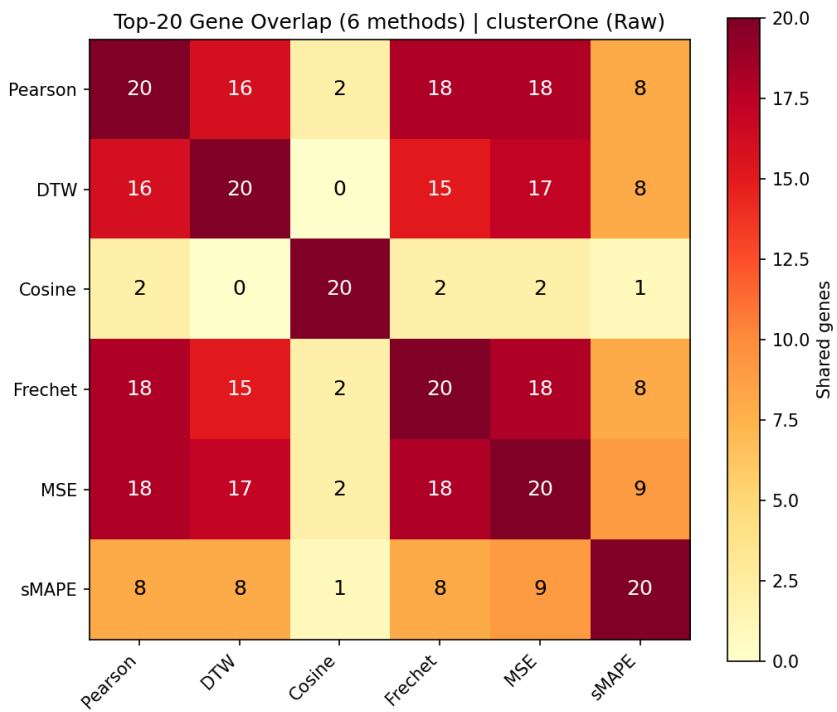
Pearson, DTW, Cosine, Fréchet, MSE all agree: 17–20/20 overlap.

sMAPE: 0/20 overlap with everything (zero-value failure at $t = 3$).

Consensus genes: TEDC1, PGAP1, TANGO6

Validates that 5 of 6 methods capture the same biology.

Results: clusterOne — Pearson/MSE/Fréchet Core

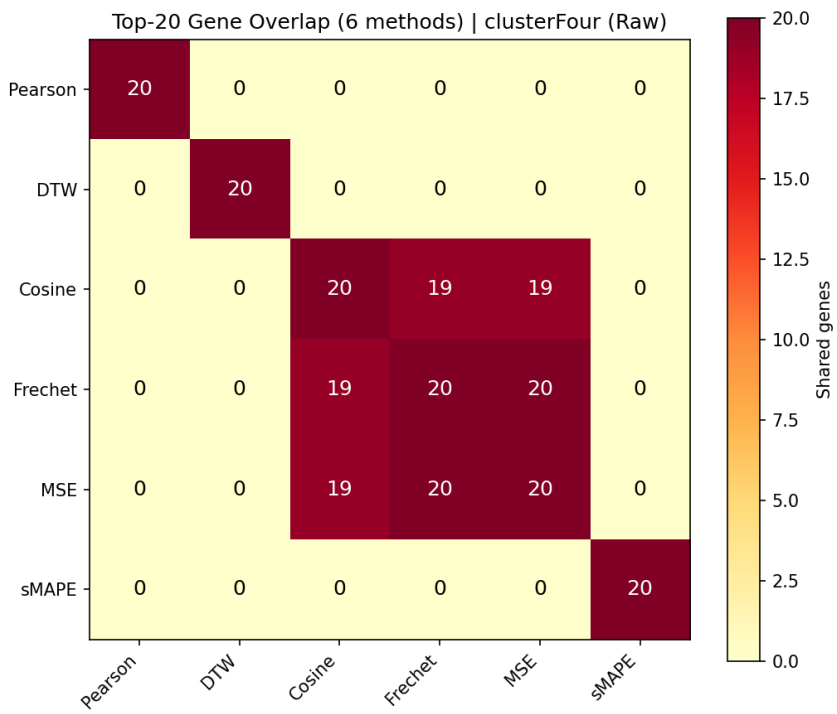


Pearson–MSE–Fréchet: 18/20 overlap
DTW joins at 15–17/20
Cosine isolated: 0–2/20

Top ensemble genes (5/6 votes):
SSR3, REXO2, YIF1A, MAZ,
ACBD3, GSPT2

Cosine finds magnitude, not shape.

Results: clusterFour — The Hard Case



70× spike ⇒ effectively a delta function.

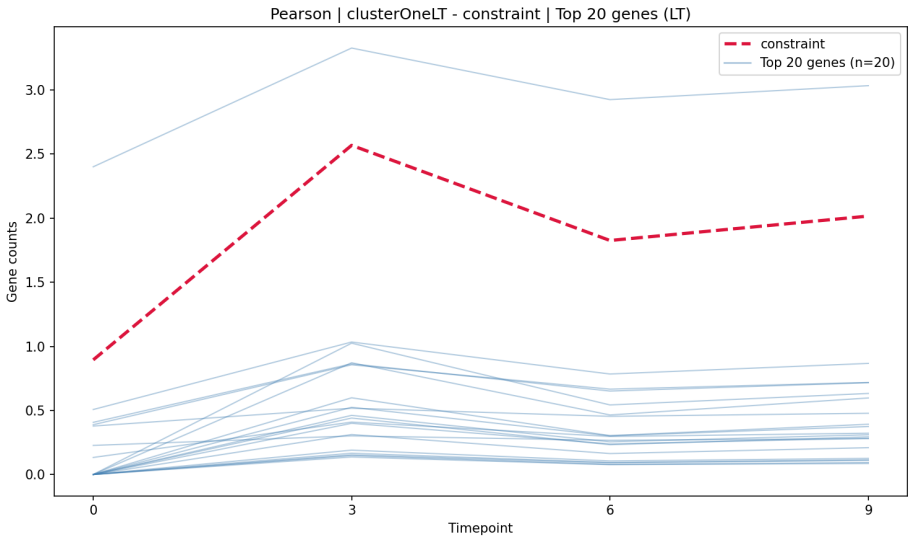
Cosine/Fréchet/MSE agree (19–20/20): all dominated by $t = 3$ magnitude.

Pearson isolated (scale-invariant, finds *relative* peaks).

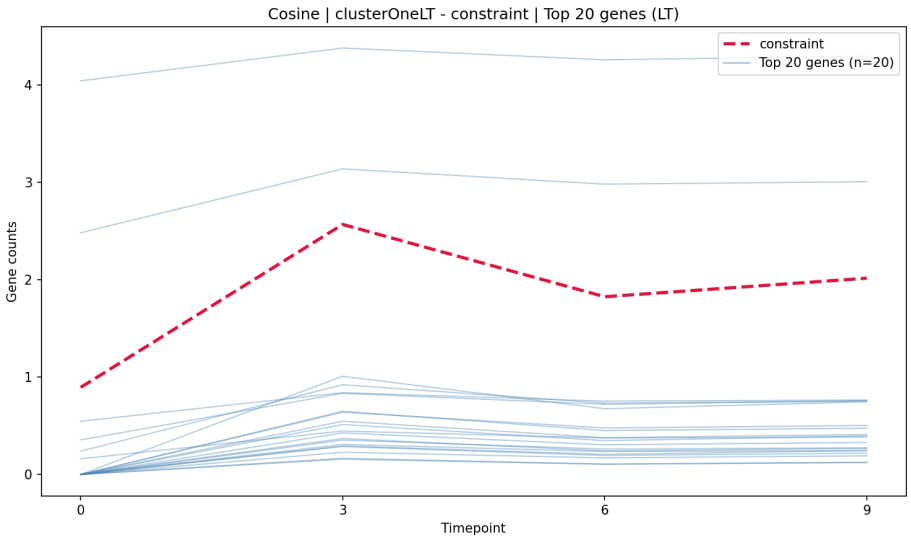
DTW, sMAPE also isolated.

No genuine shape consensus.
Needs denser time sampling to resolve.

Pearson vs Cosine — Visual Comparison (clusterOne LT)



Pearson: shape-matching genes



Cosine: magnitude-biased genes

Ensemble Rankings — clusterTwo (Raw)

Rank	Gene	Votes	Methods
1	FAM49B	**6/6**	All methods
2	RNF213	**6/6**	All methods
3	ARGLU1	4/6	Pearson, DTW, Fréchet, MSE
4	TMC8	5/6	Pearson, Cosine, Fréchet, MSE, sMAPE
5	SNRNP70	5/6	Pearson, Cosine, Fréchet, MSE, sMAPE

FAM49B and RNF213 are unanimous across all 6 methods — highest confidence candidates for TCR pathway involvement.

Recommendations

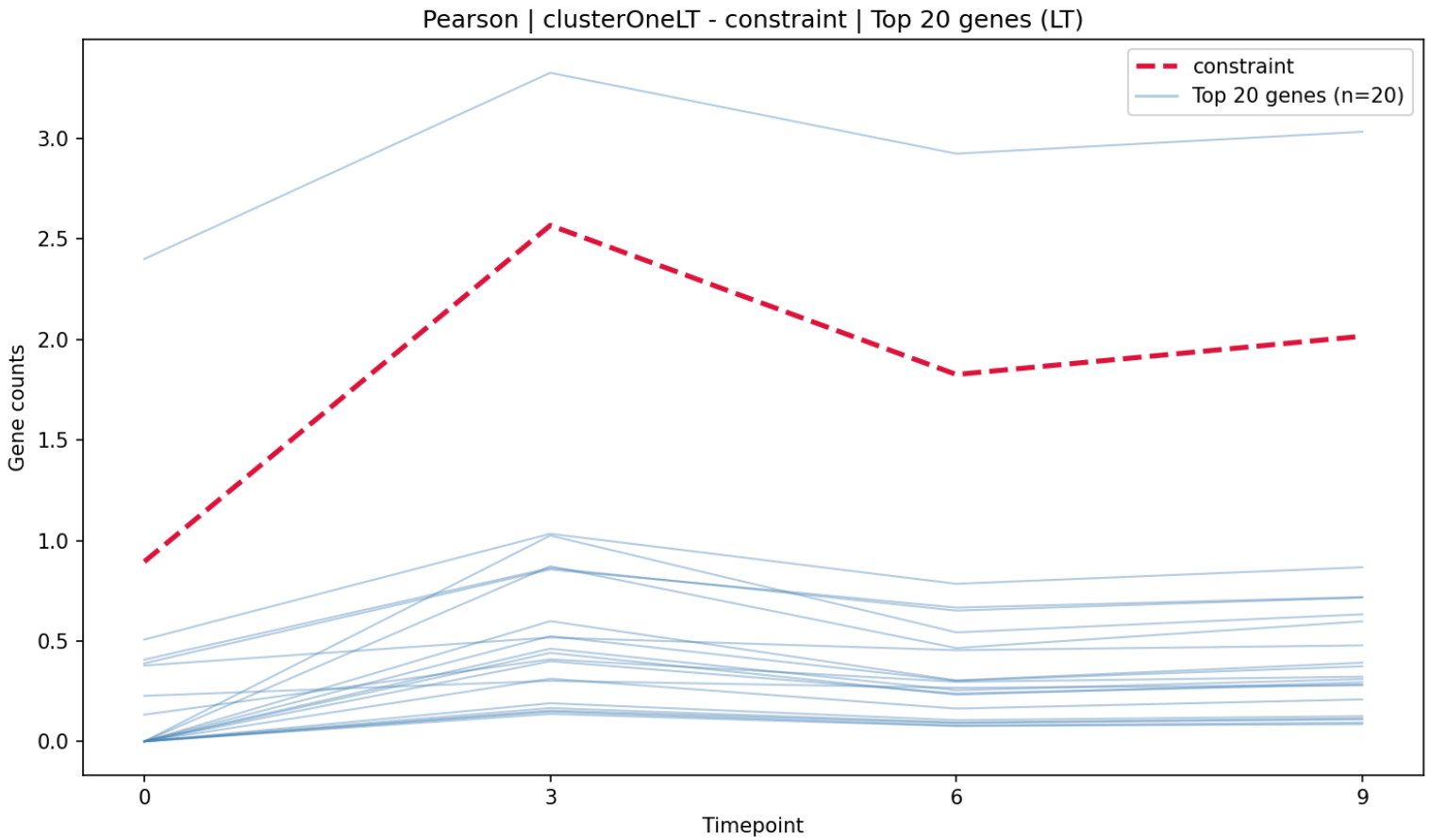
1. **Primary metric:** Pearson — directly captures shape, invariant to scale and baseline
2. **Secondary validation:** MSE — independent distance-based confirmation (Fréchet is near-redundant with MSE)
3. **Skip for this data:** Cosine (inconsistent), sMAPE (zero-point failure)
4. **DTW:** marginal benefit at 4 timepoints; valuable with denser sampling
5. **Ensemble:** 3-method consensus (Pearson + DTW + MSE), $\geq 2/3$ agreement for high-confidence gene lists
6. **clusterFour:** needs more timepoints or peak-detection approach

Thank you

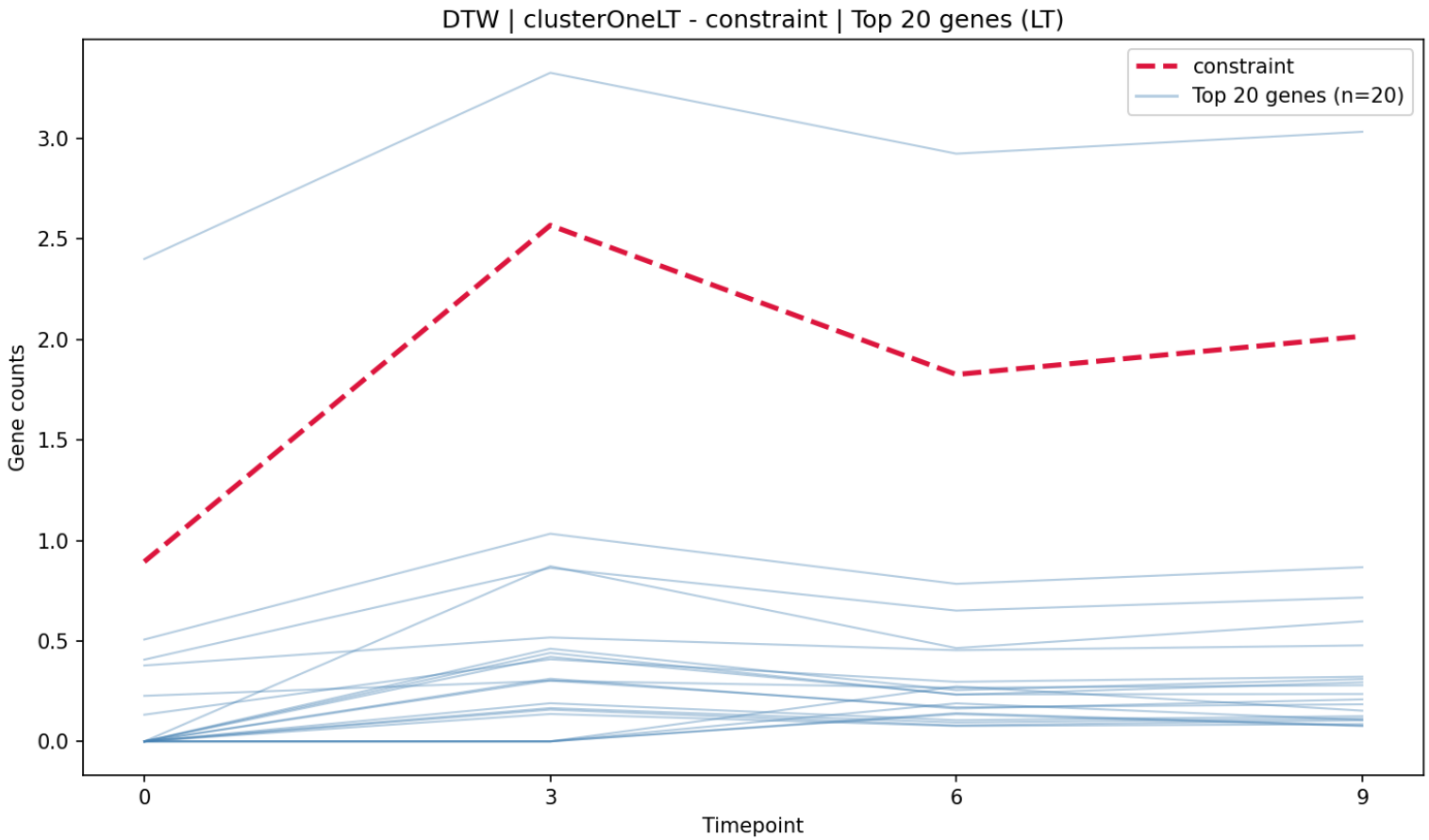
Appendix

All Methods — Log-Transformed Clusters

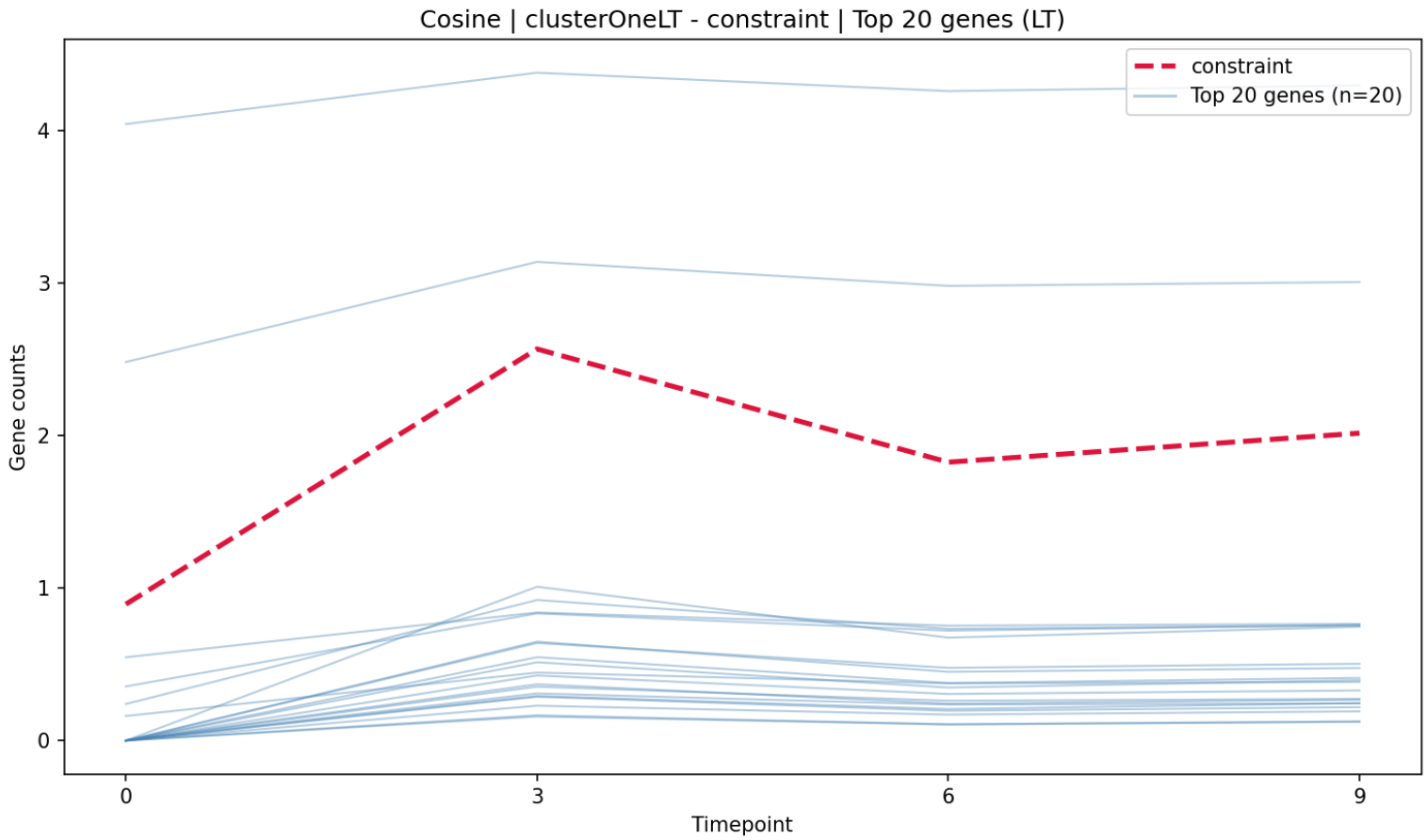
clusterOne LT — Pearson



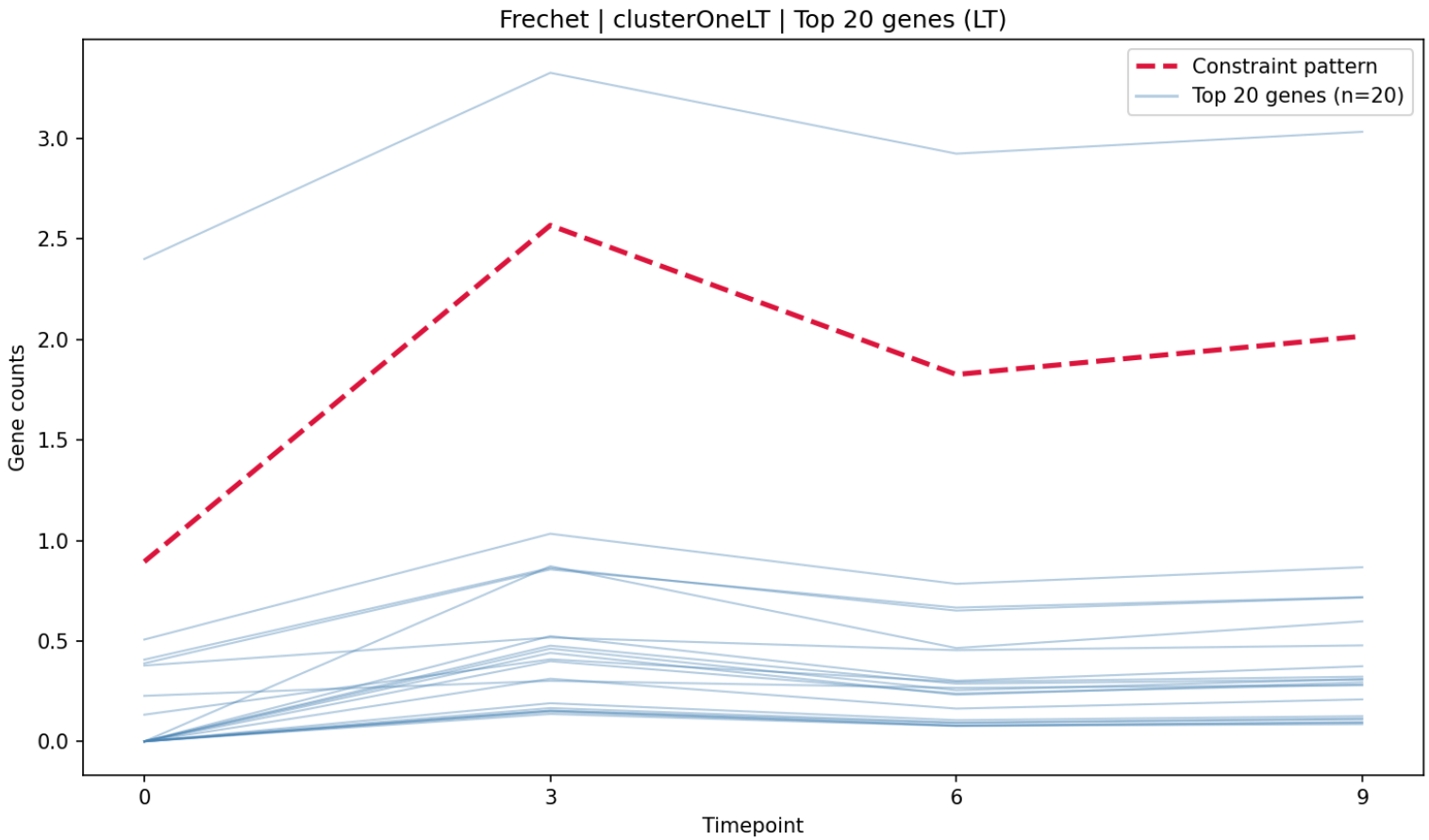
clusterOne LT — DTW



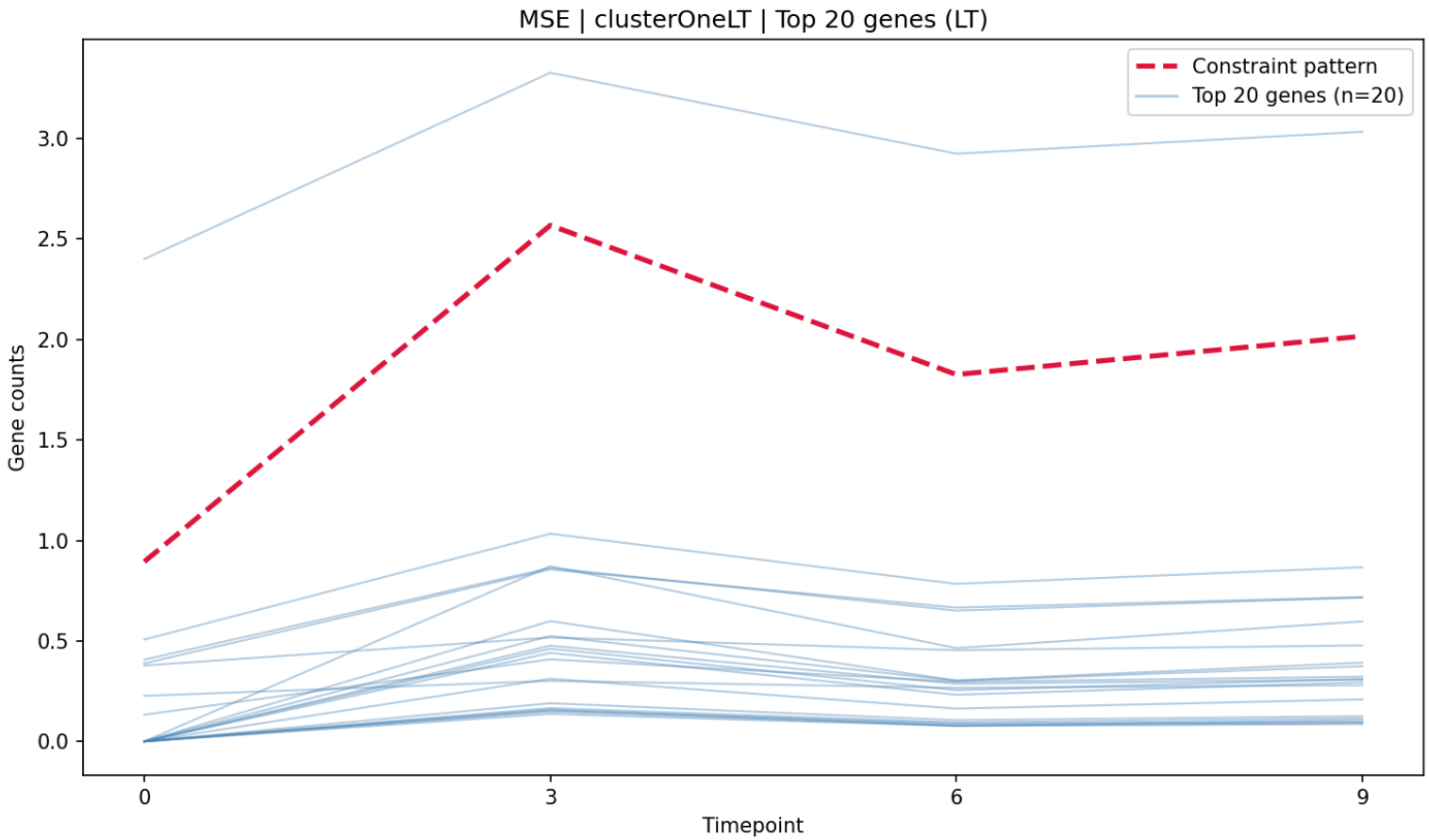
clusterOne LT — Cosine



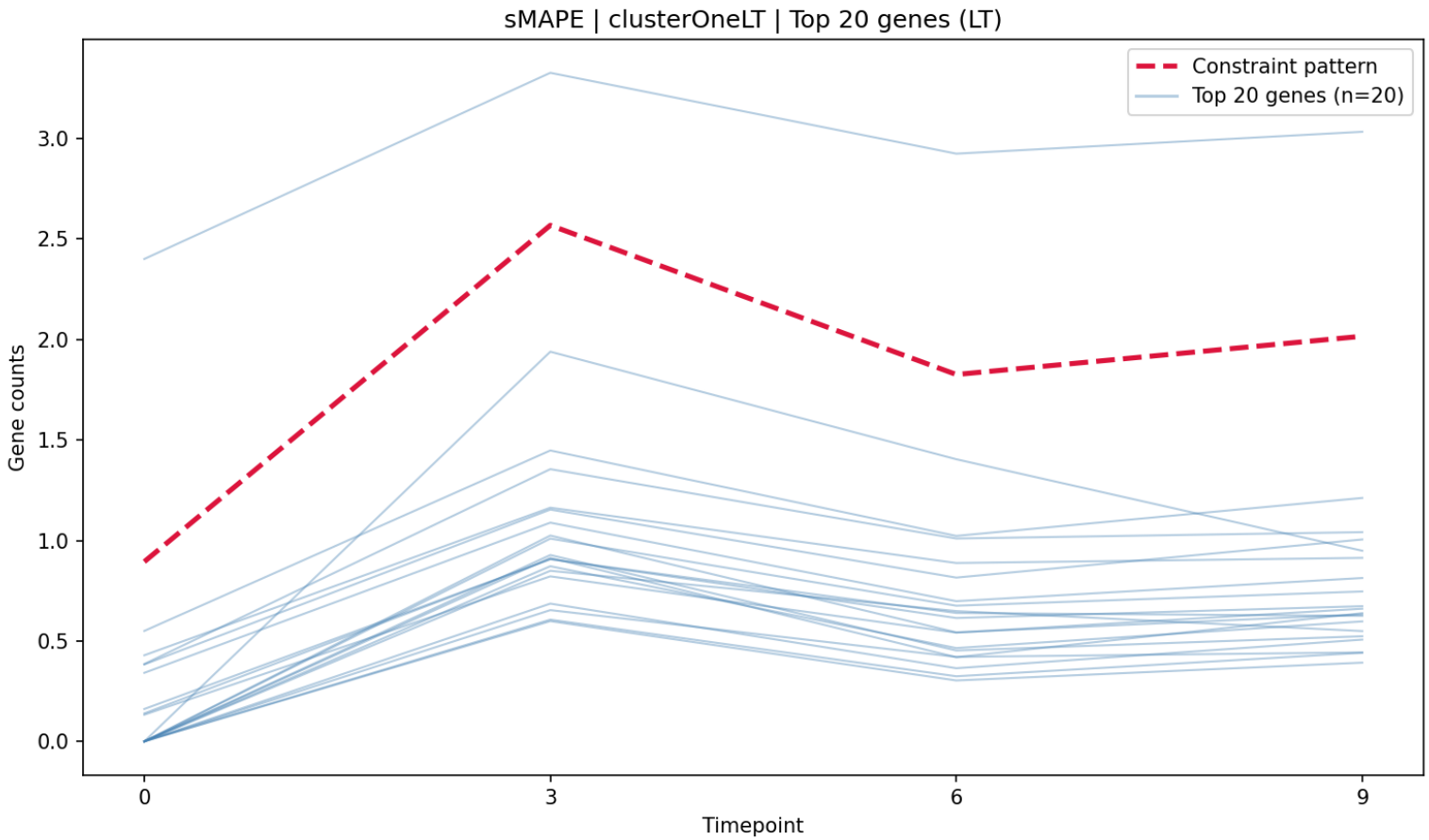
clusterOne LT — Fréchet



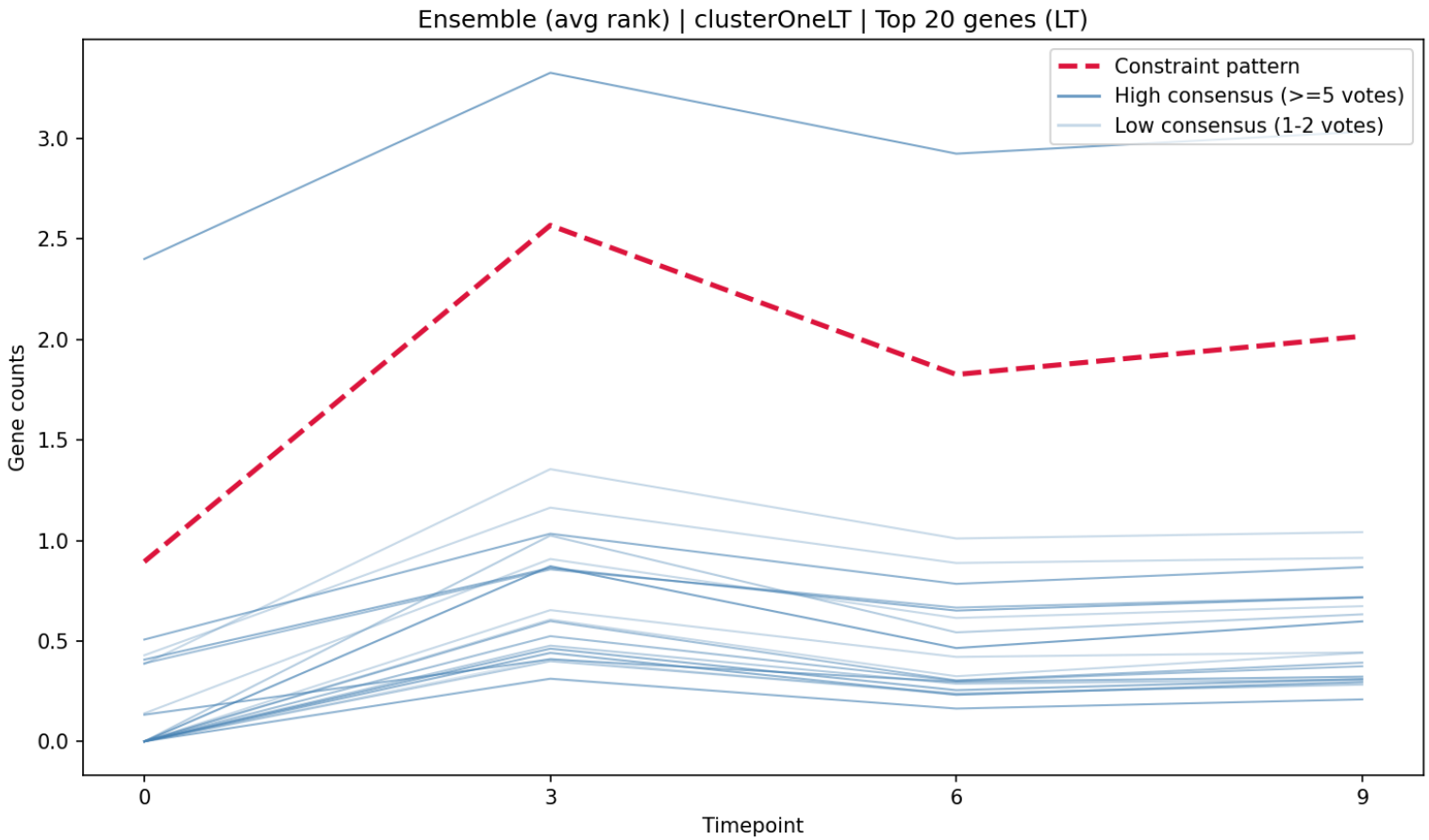
clusterOne LT — MSE



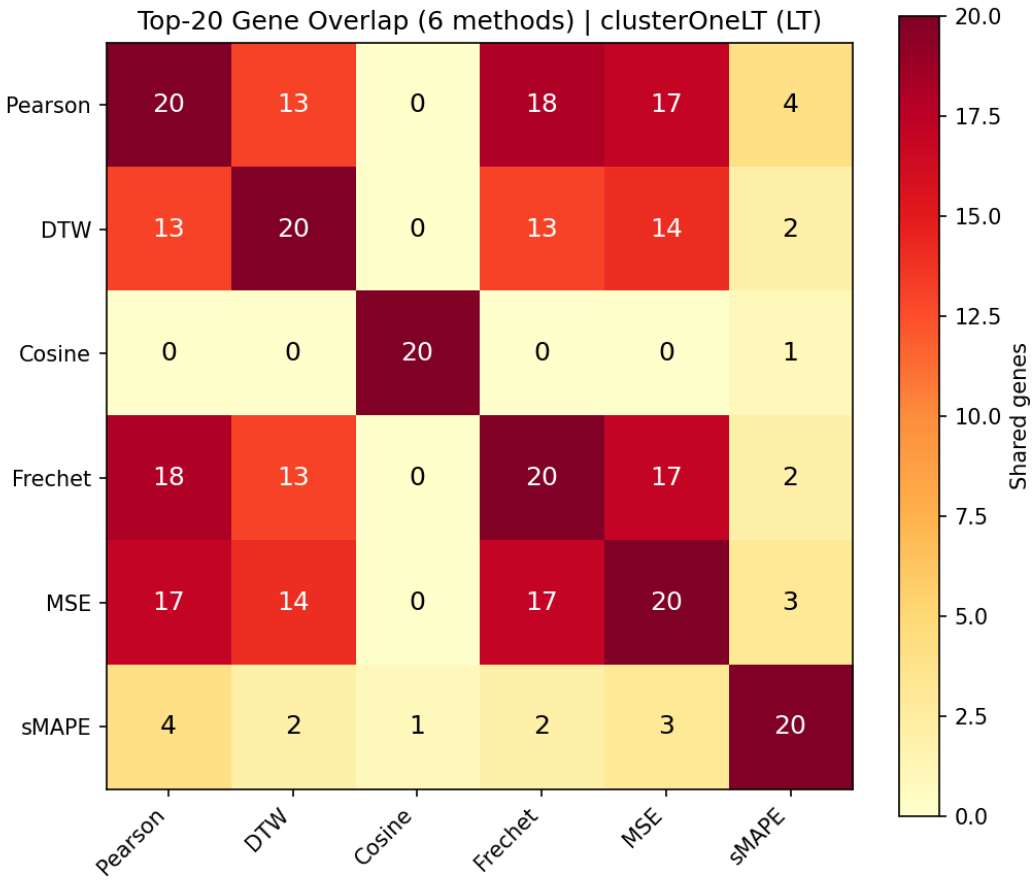
clusterOne LT — sMAPE



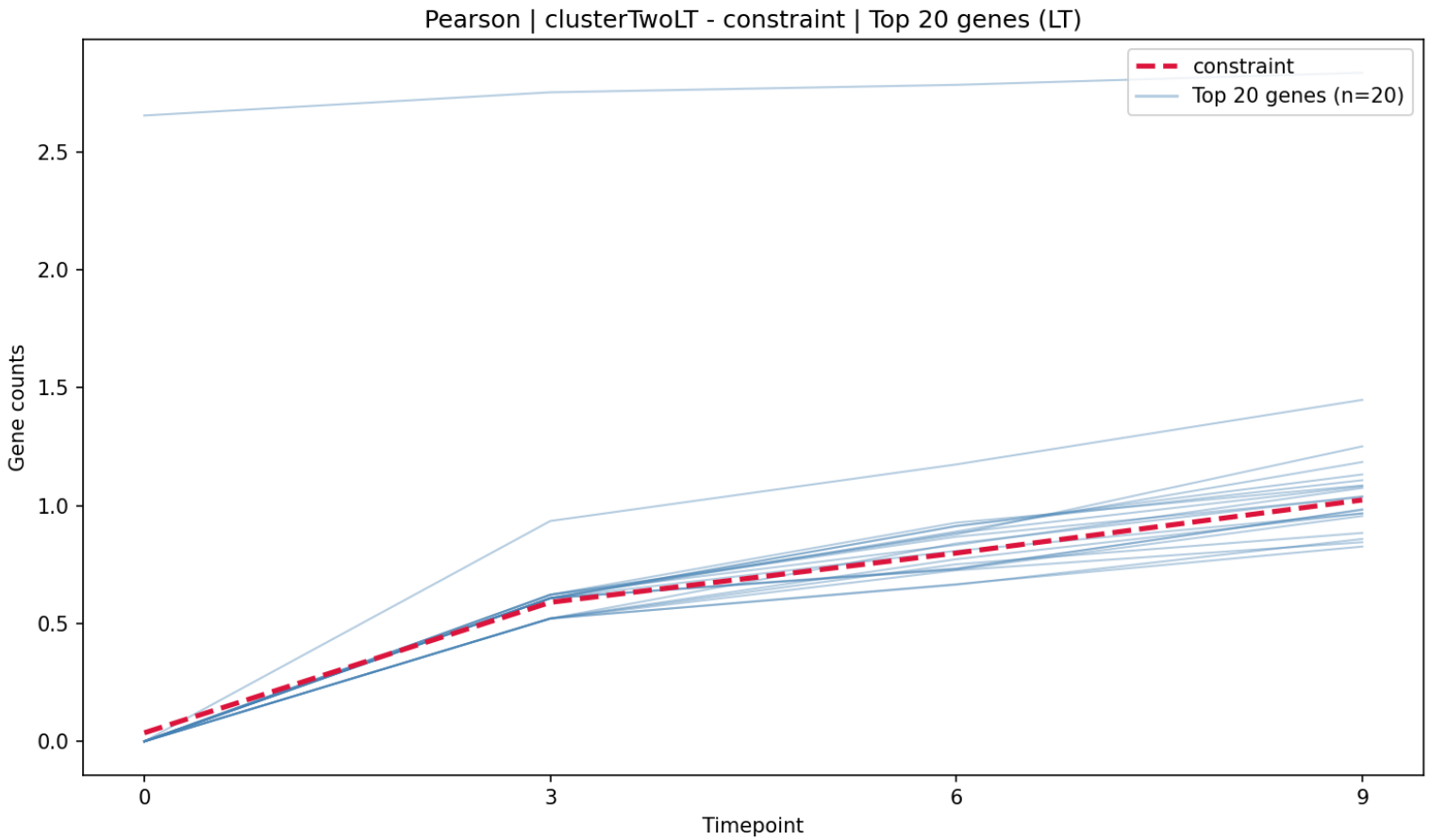
clusterOne LT — Ensemble



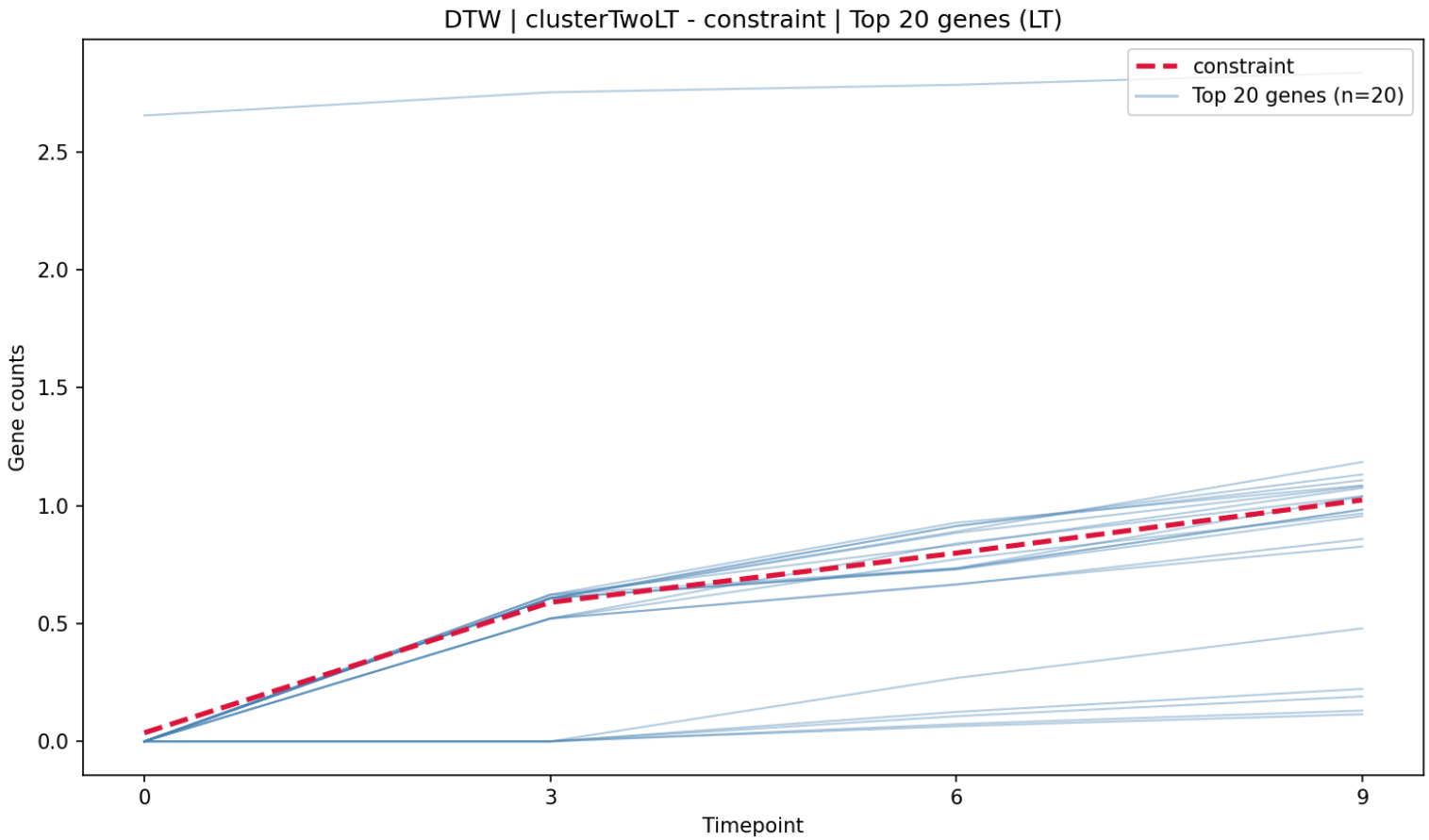
clusterOne LT — Overlap Heatmap



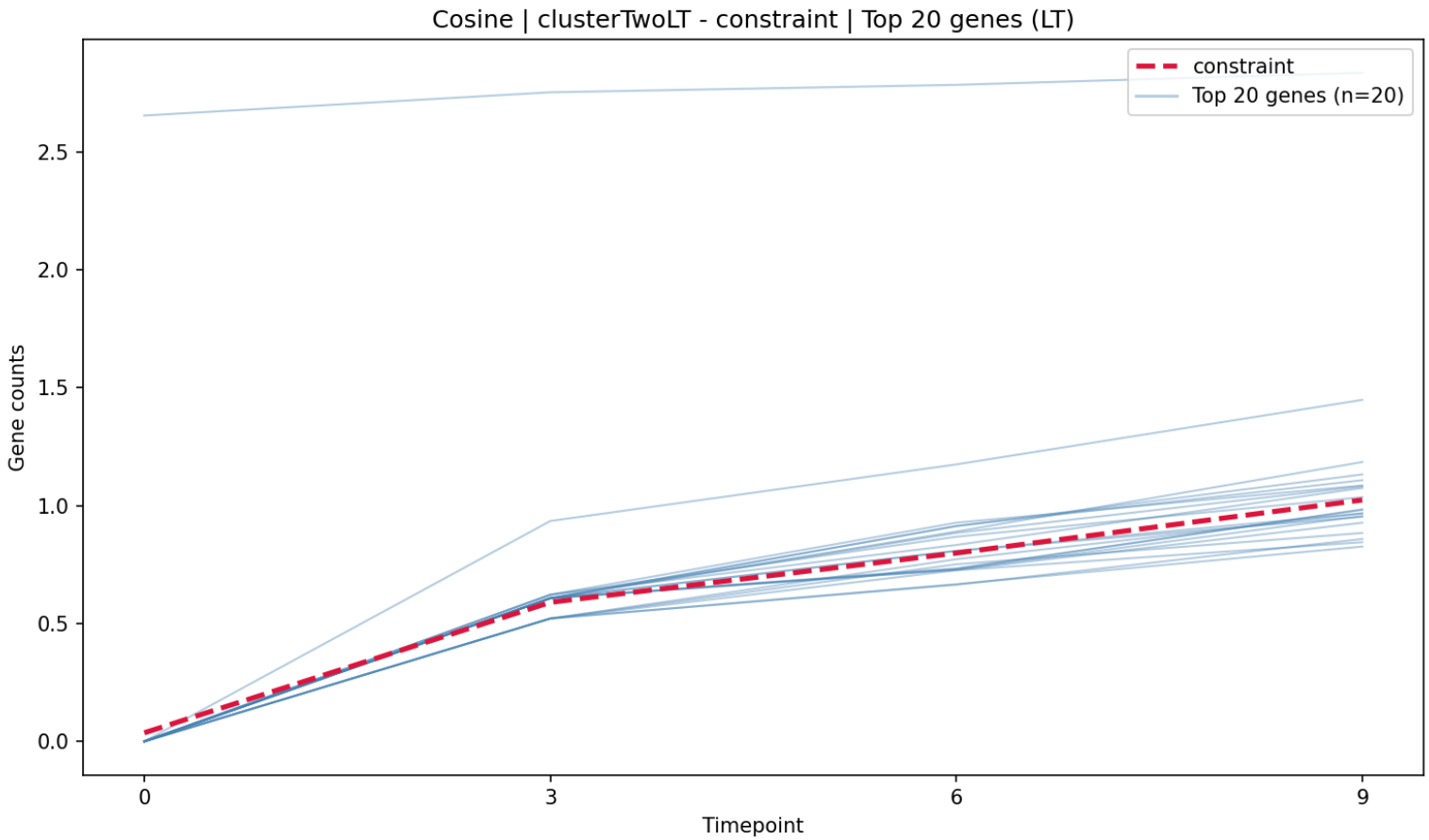
clusterTwo LT — Pearson



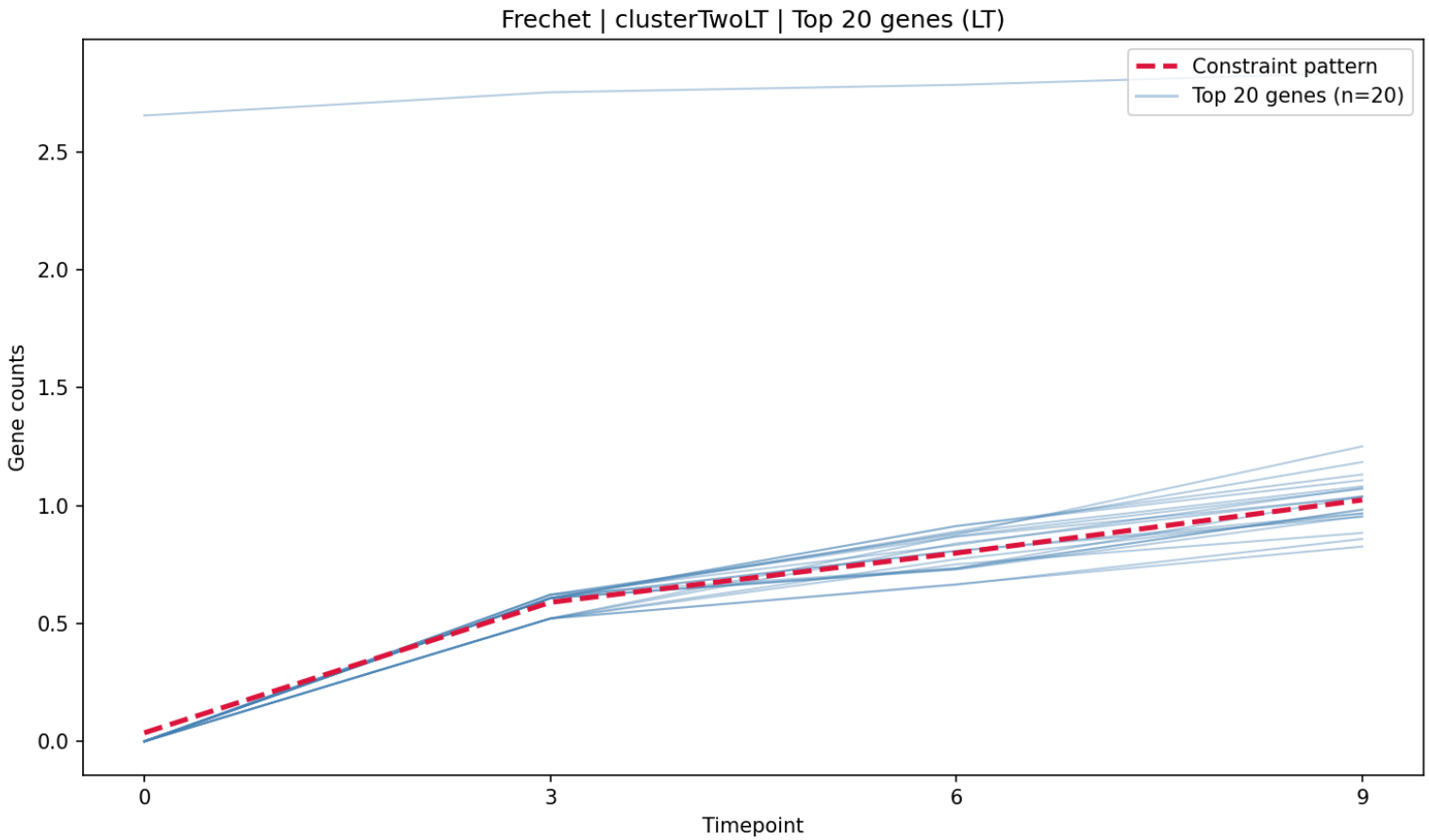
clusterTwo LT — DTW



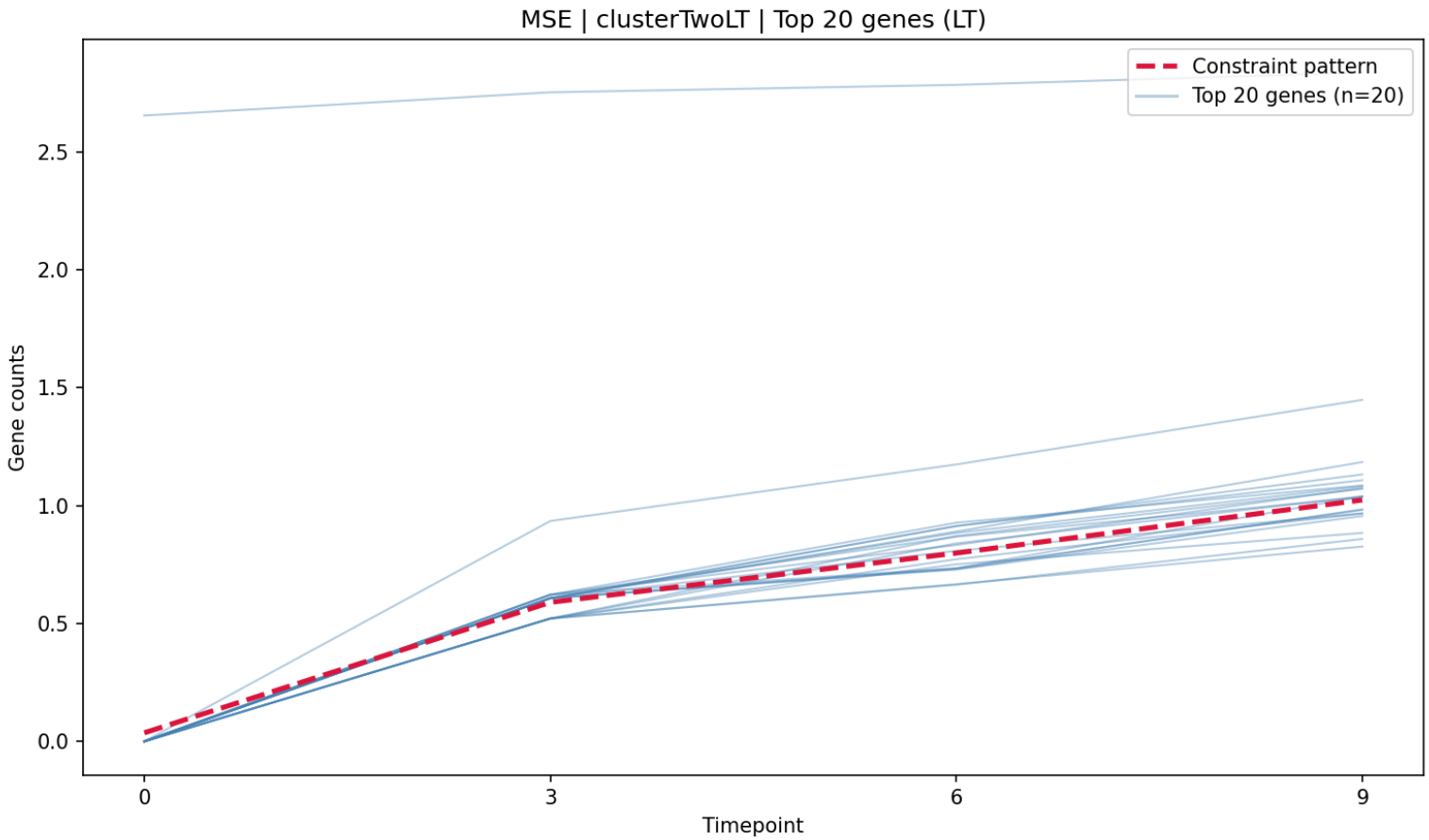
clusterTwo LT — Cosine



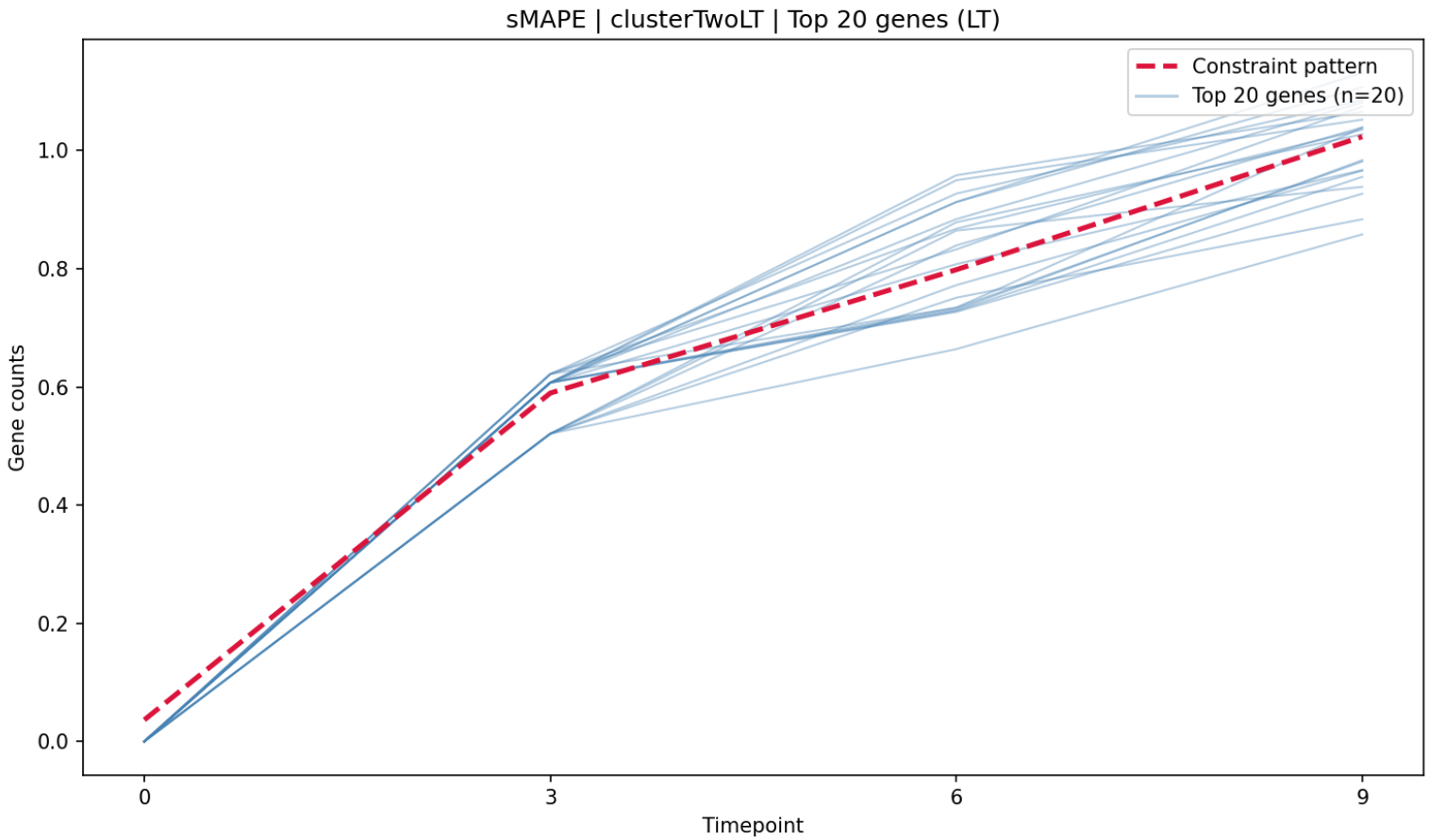
clusterTwo LT — Fréchet



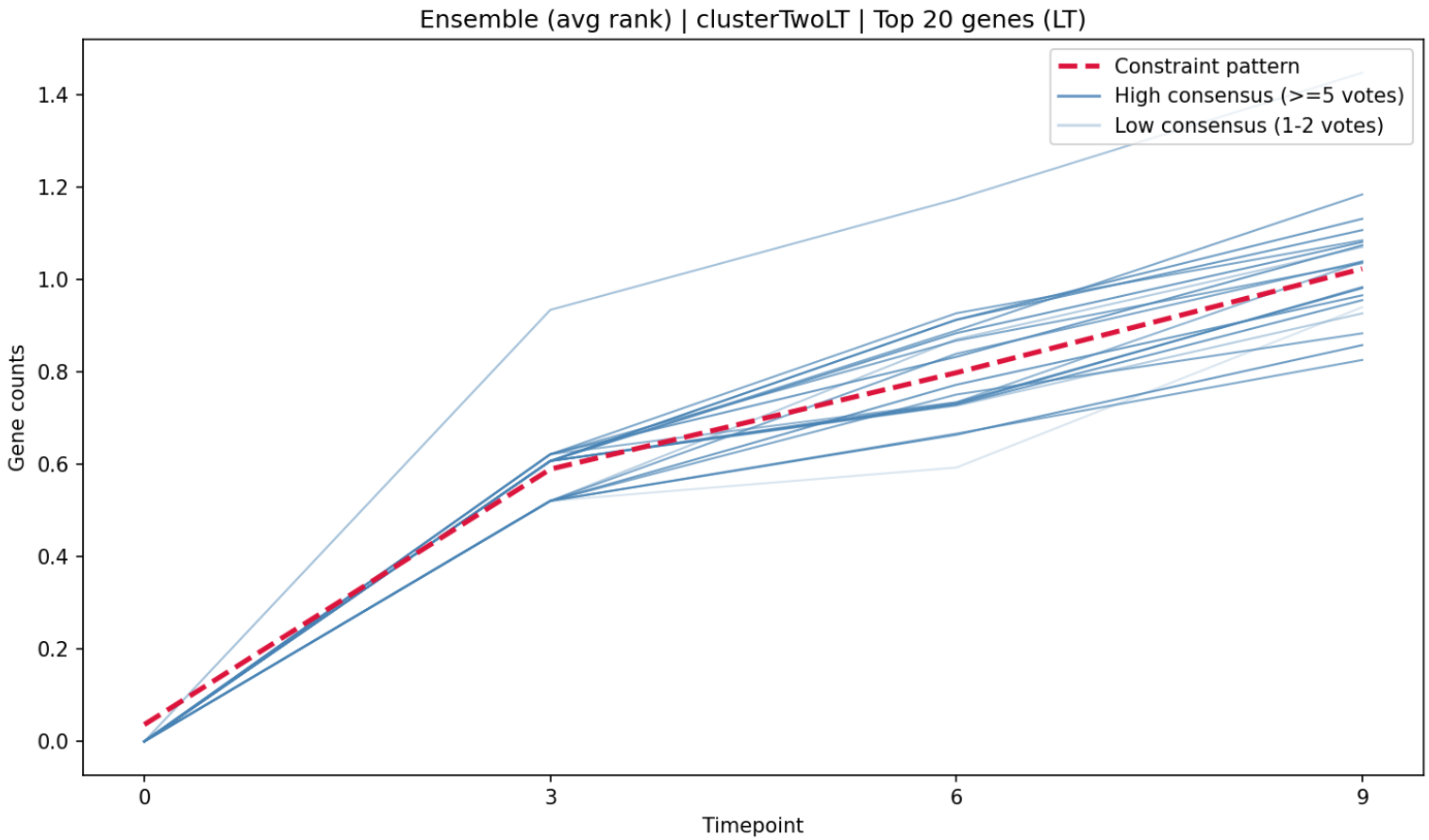
clusterTwo LT — MSE



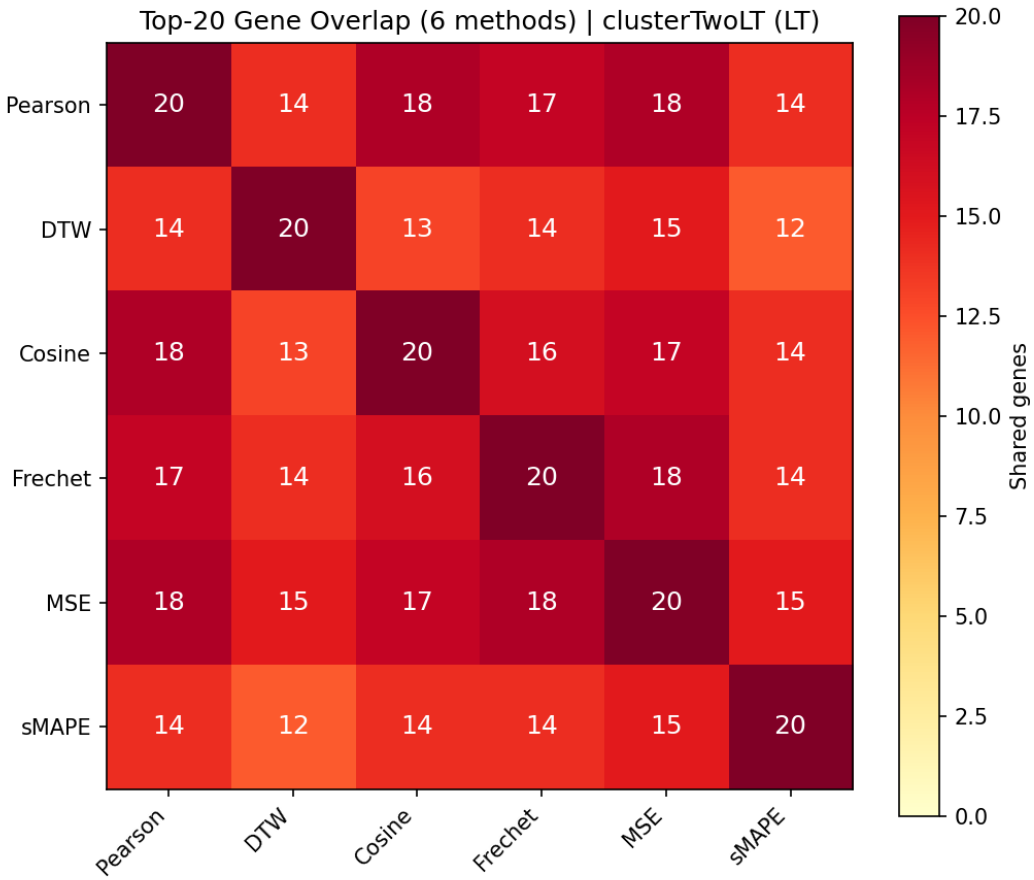
clusterTwo LT — sMAPE



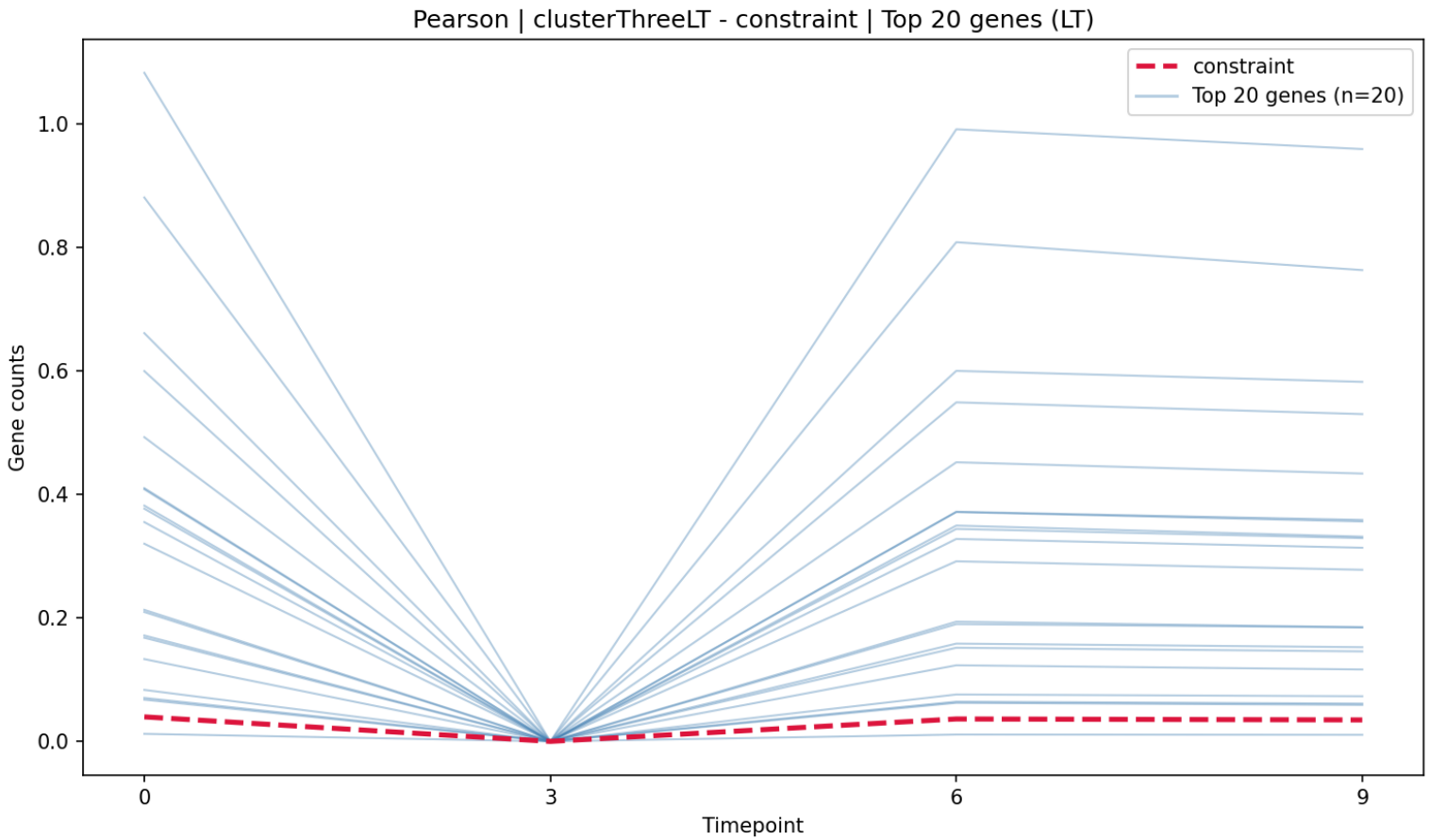
clusterTwo LT — Ensemble



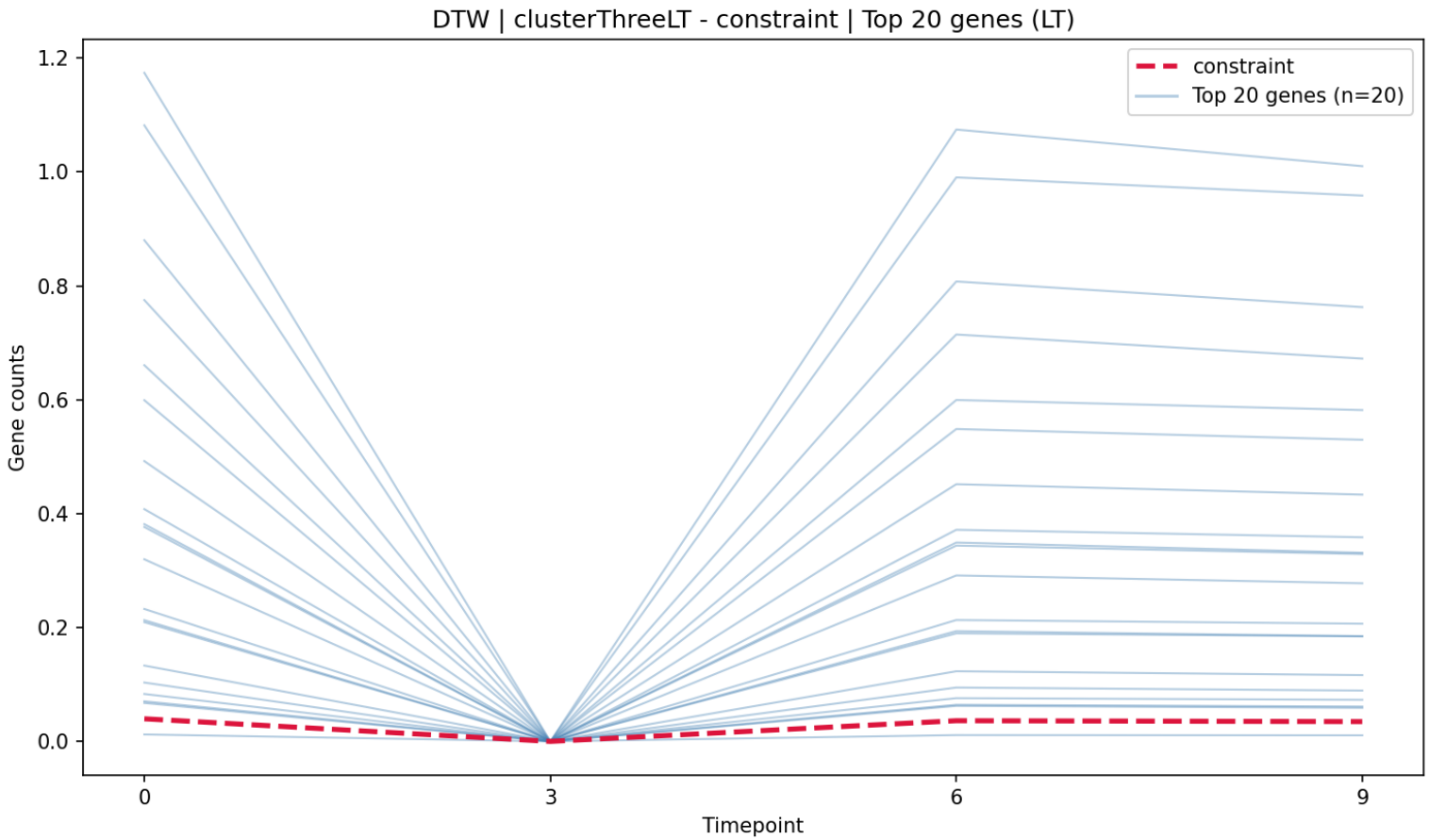
clusterTwo LT — Overlap Heatmap



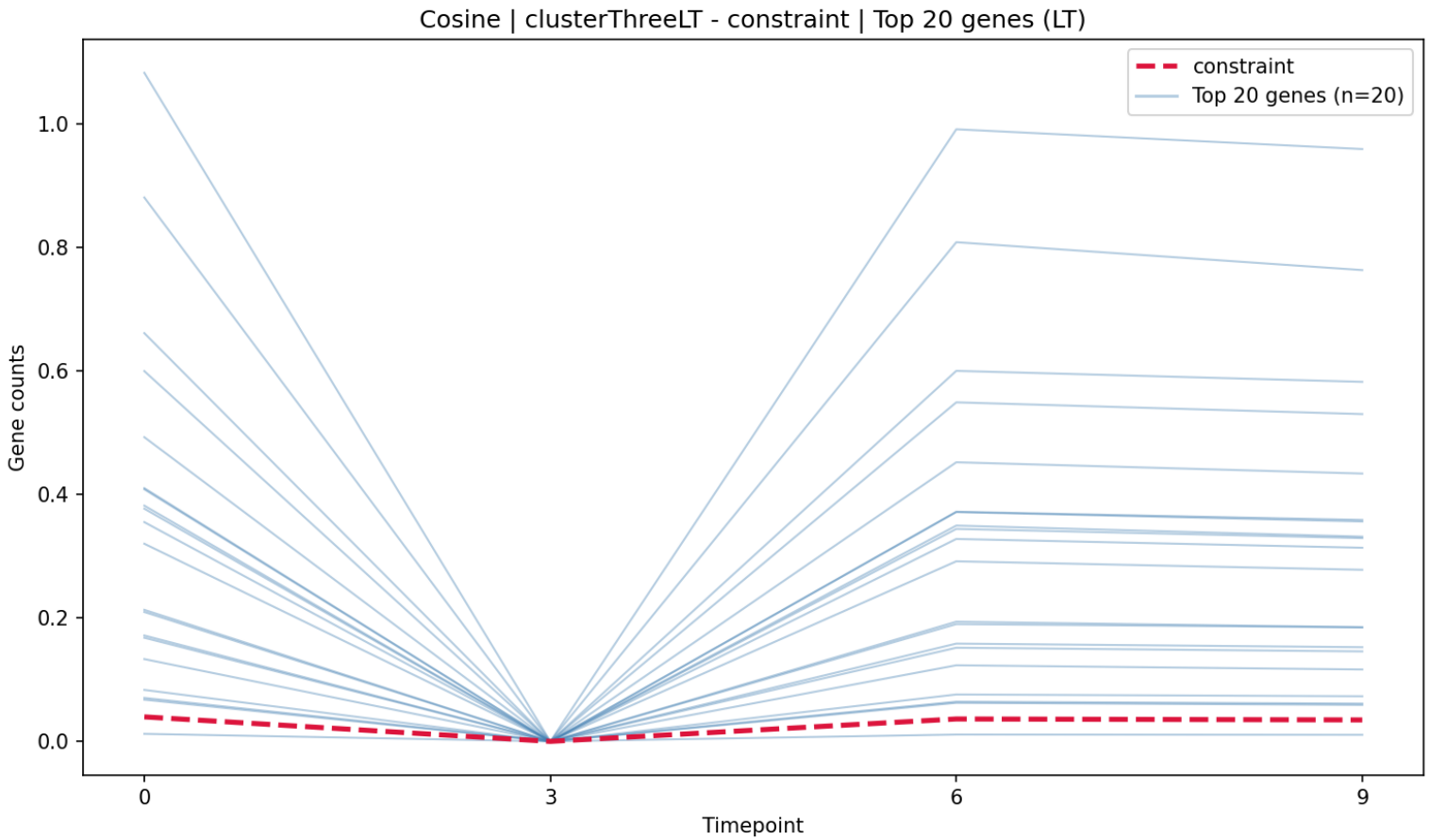
clusterThree LT — Pearson



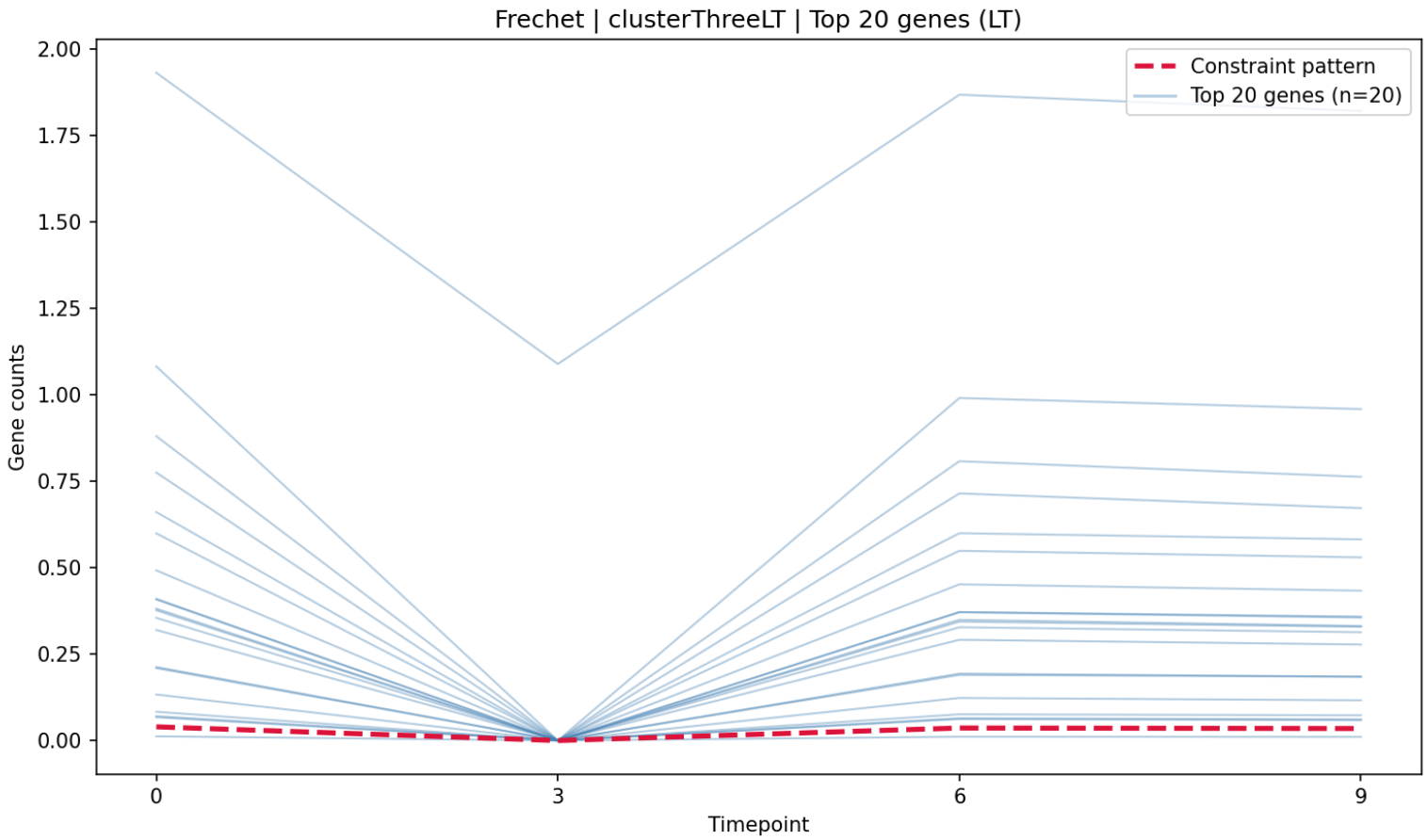
clusterThree LT — DTW



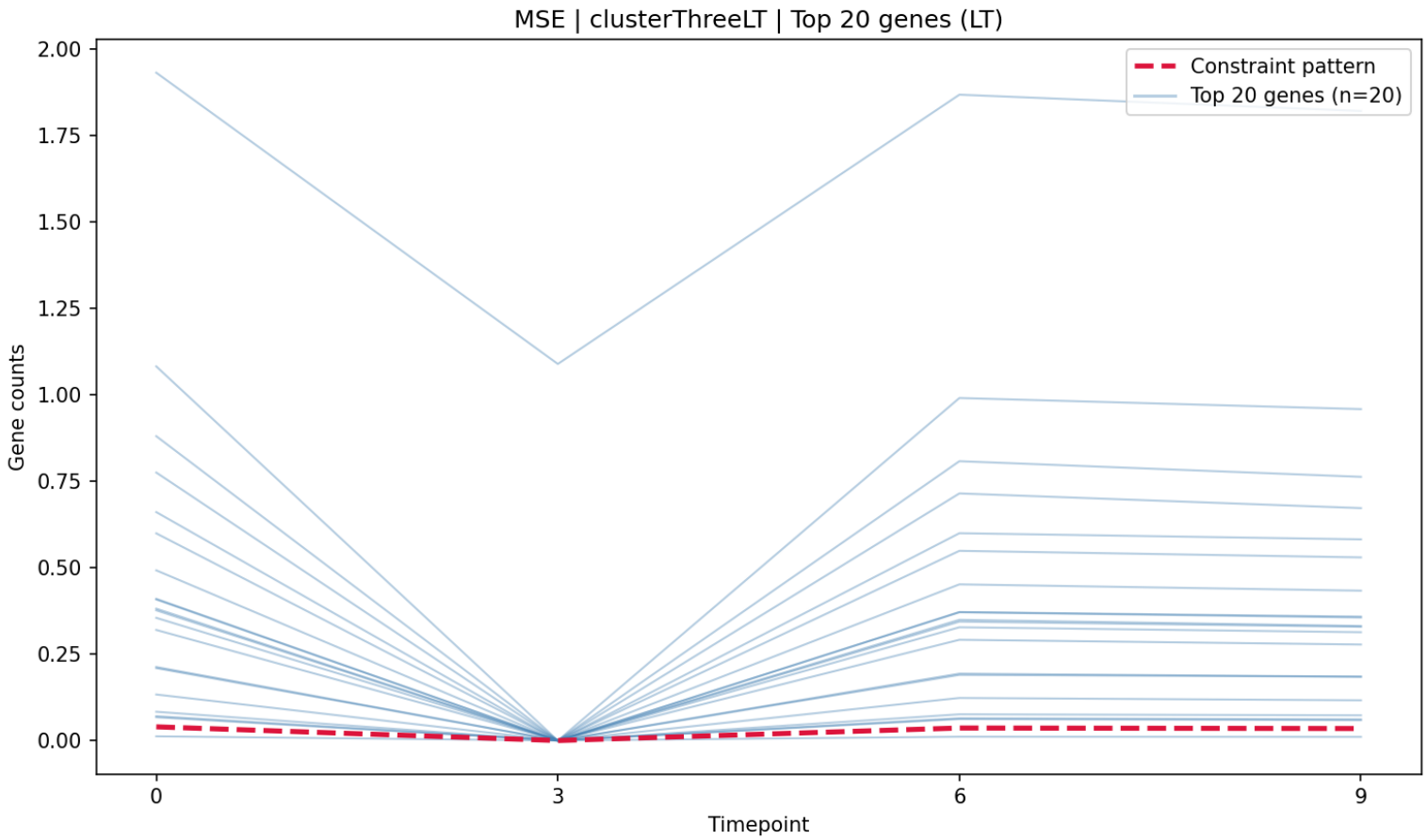
clusterThree LT — Cosine



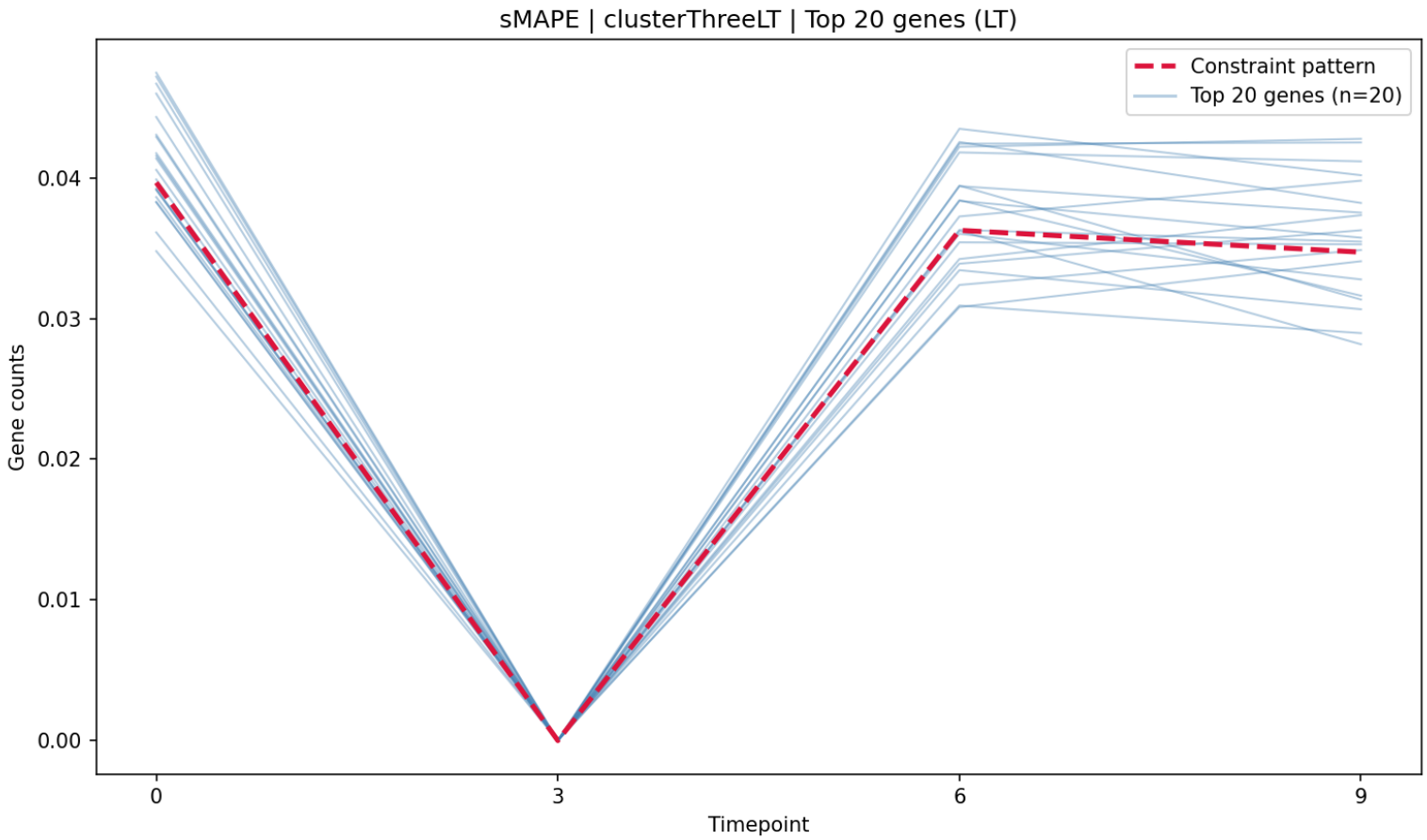
clusterThree LT — Fréchet



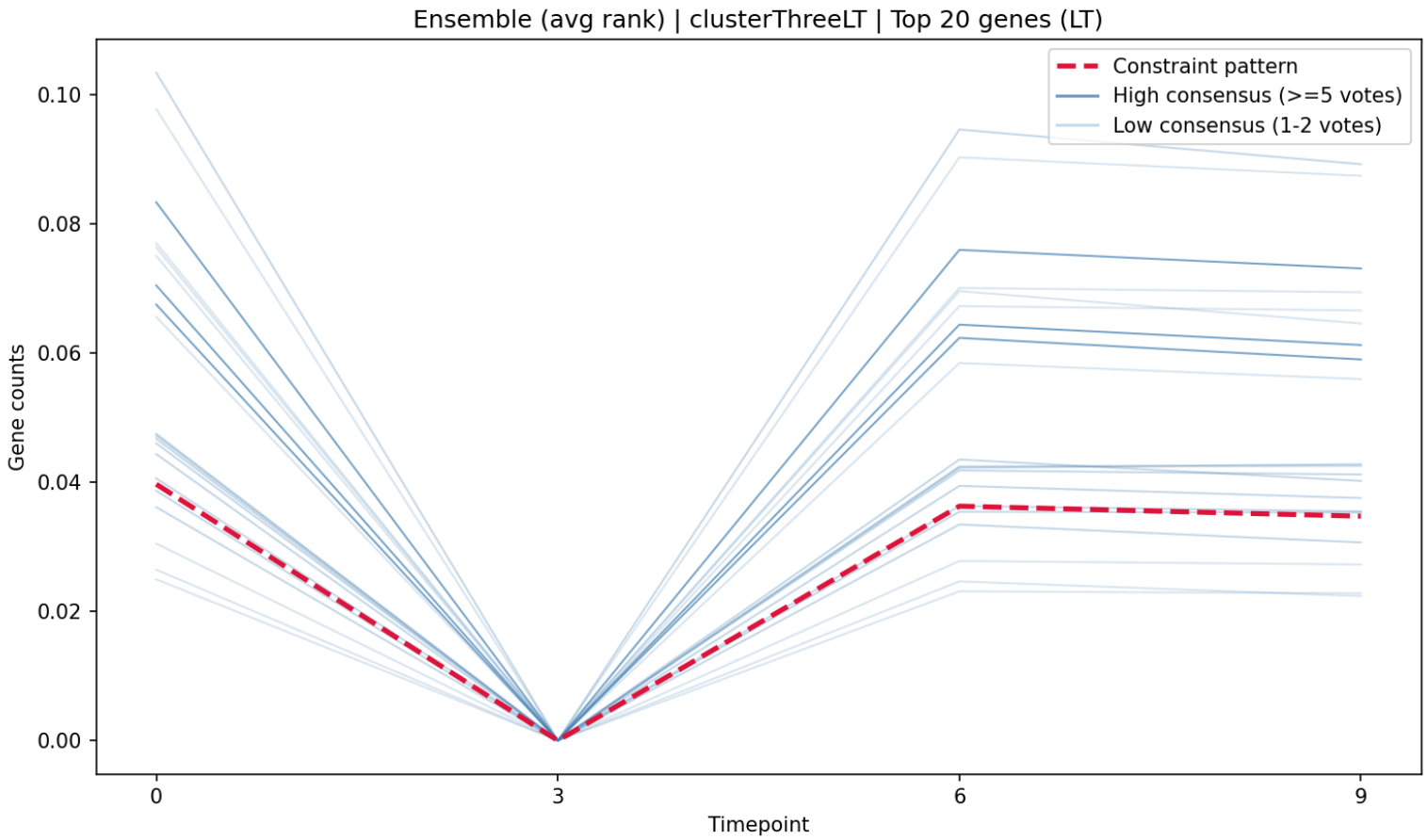
clusterThree LT — MSE



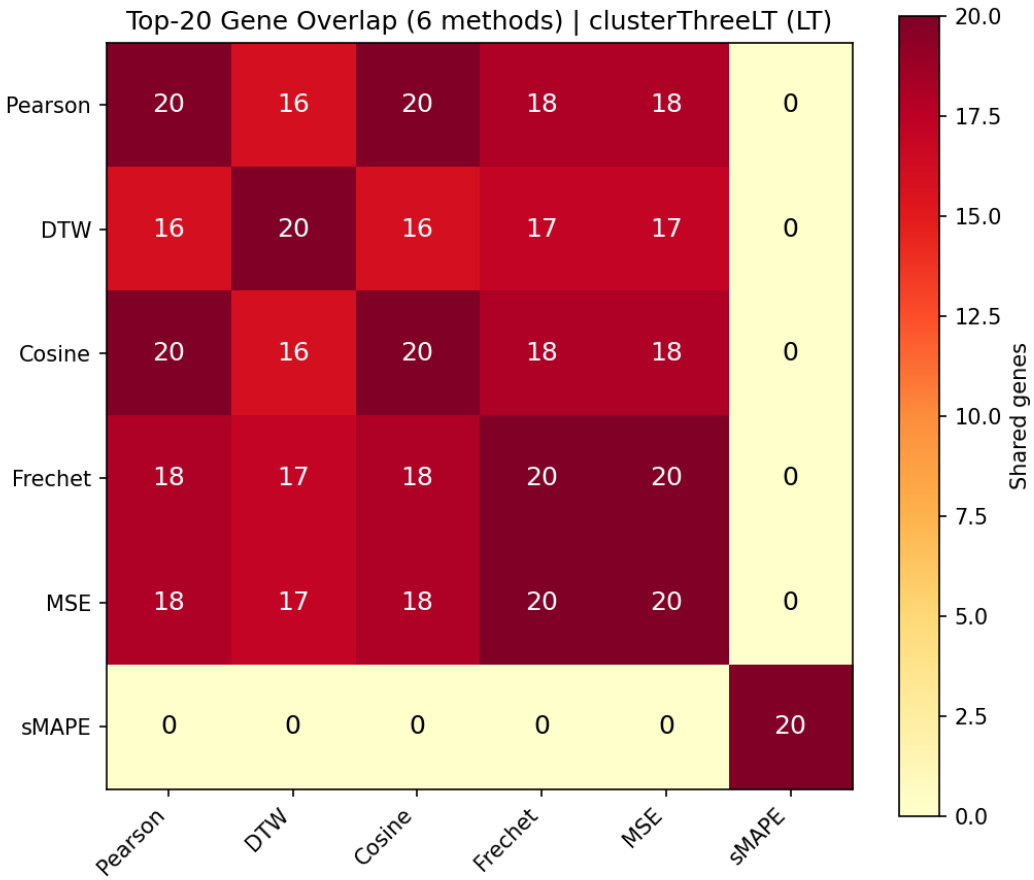
clusterThree LT — sMAPE



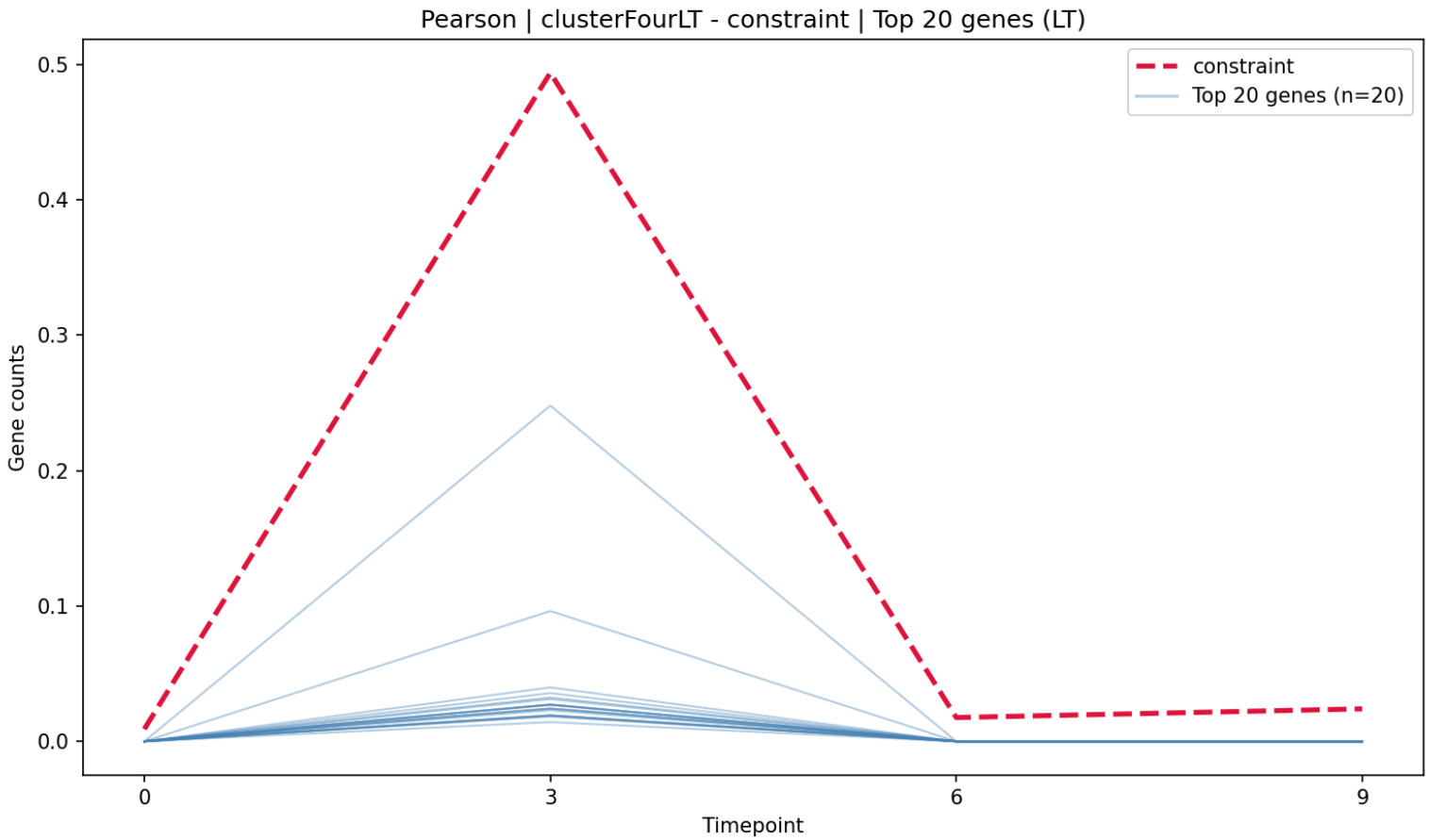
clusterThree LT — Ensemble



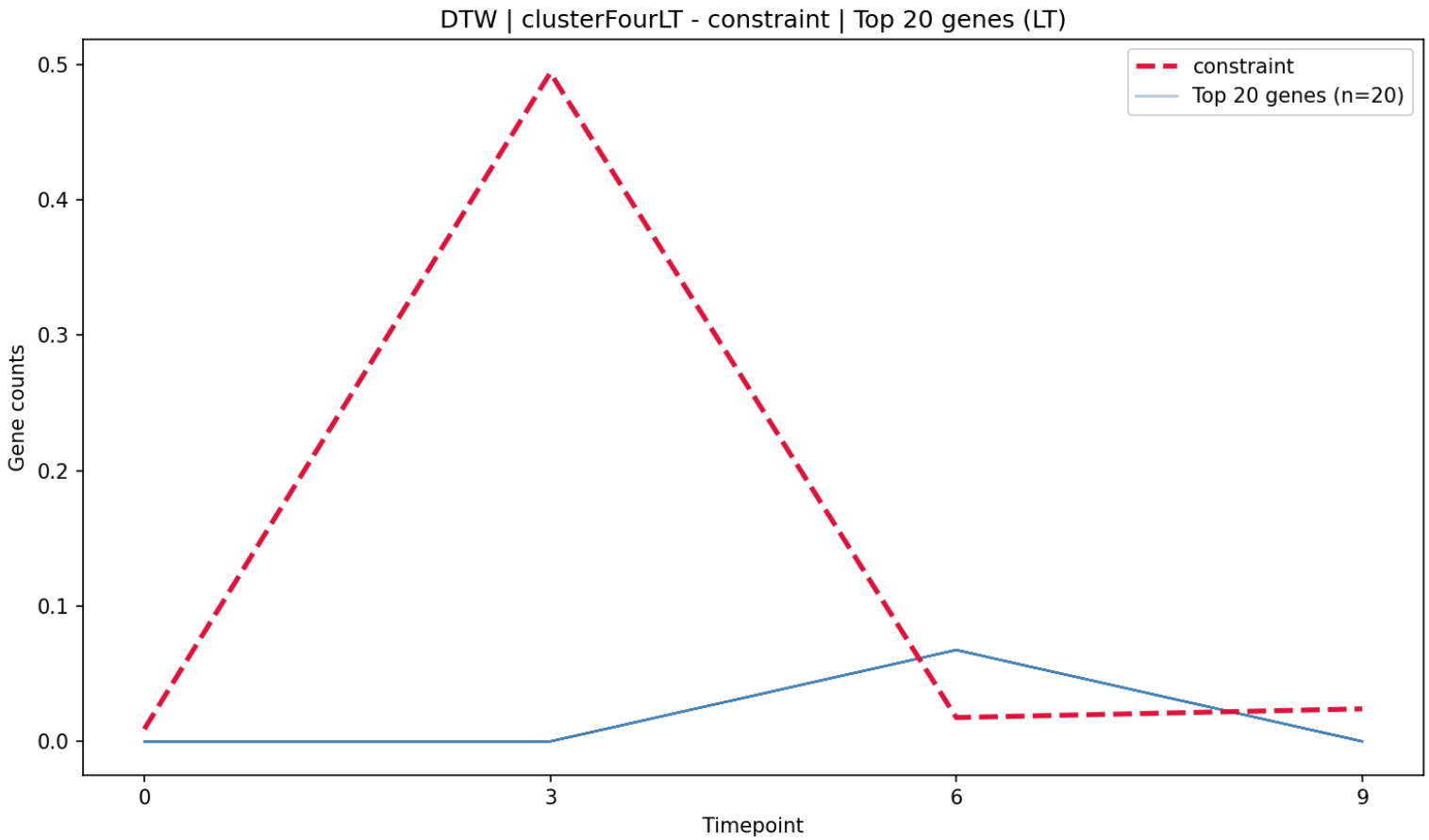
clusterThree LT — Overlap Heatmap



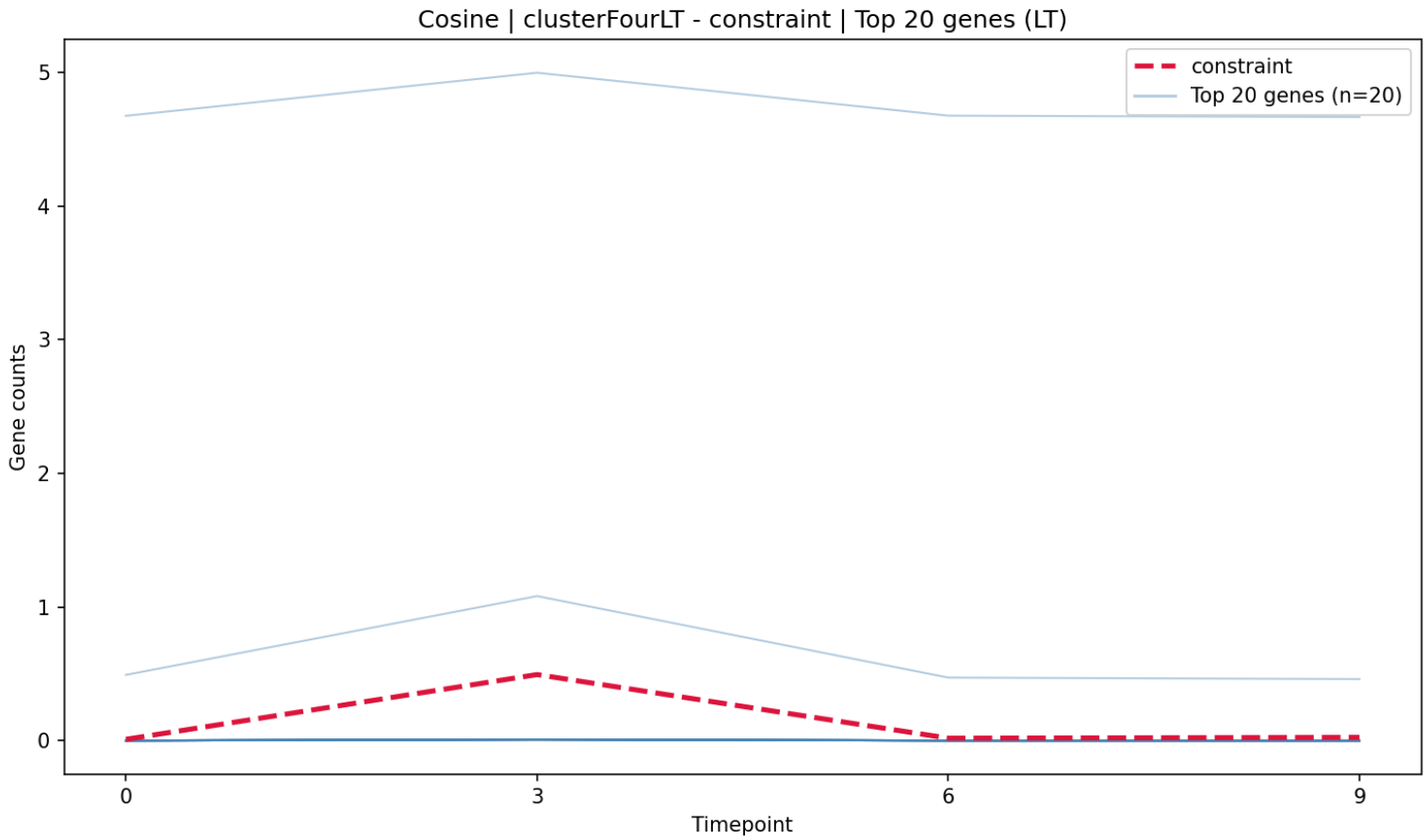
clusterFour LT — Pearson



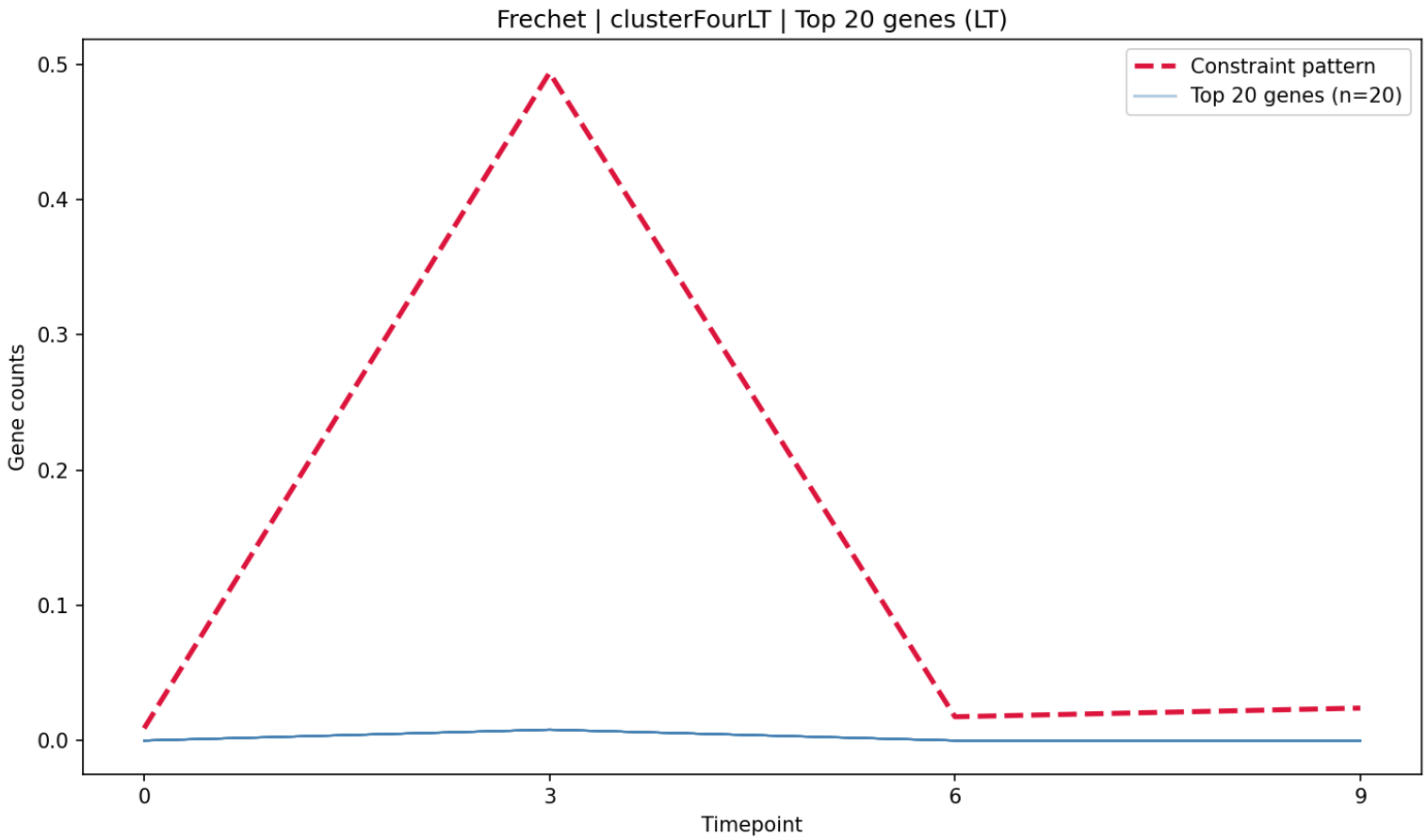
clusterFour LT — DTW



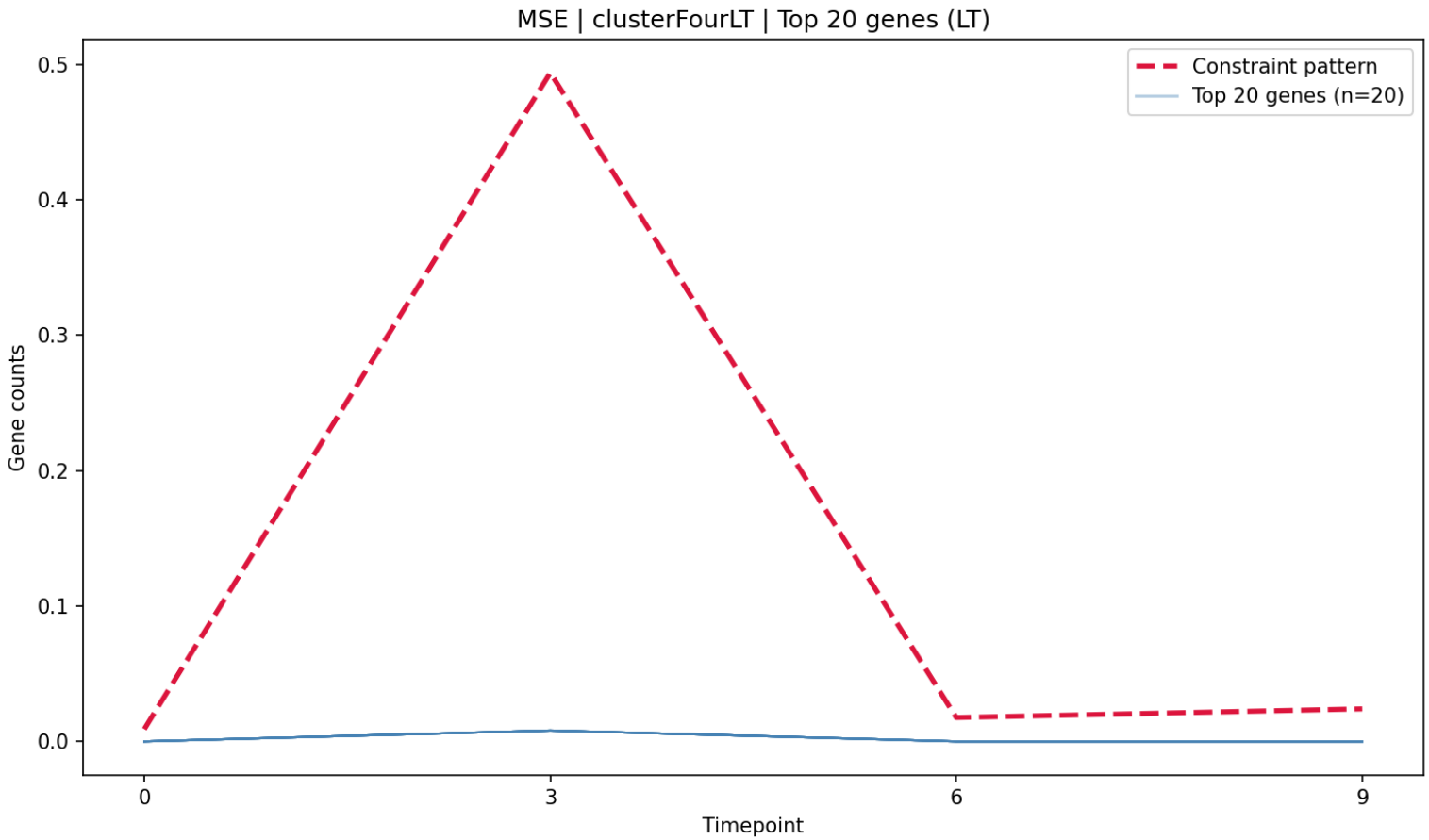
clusterFour LT — Cosine



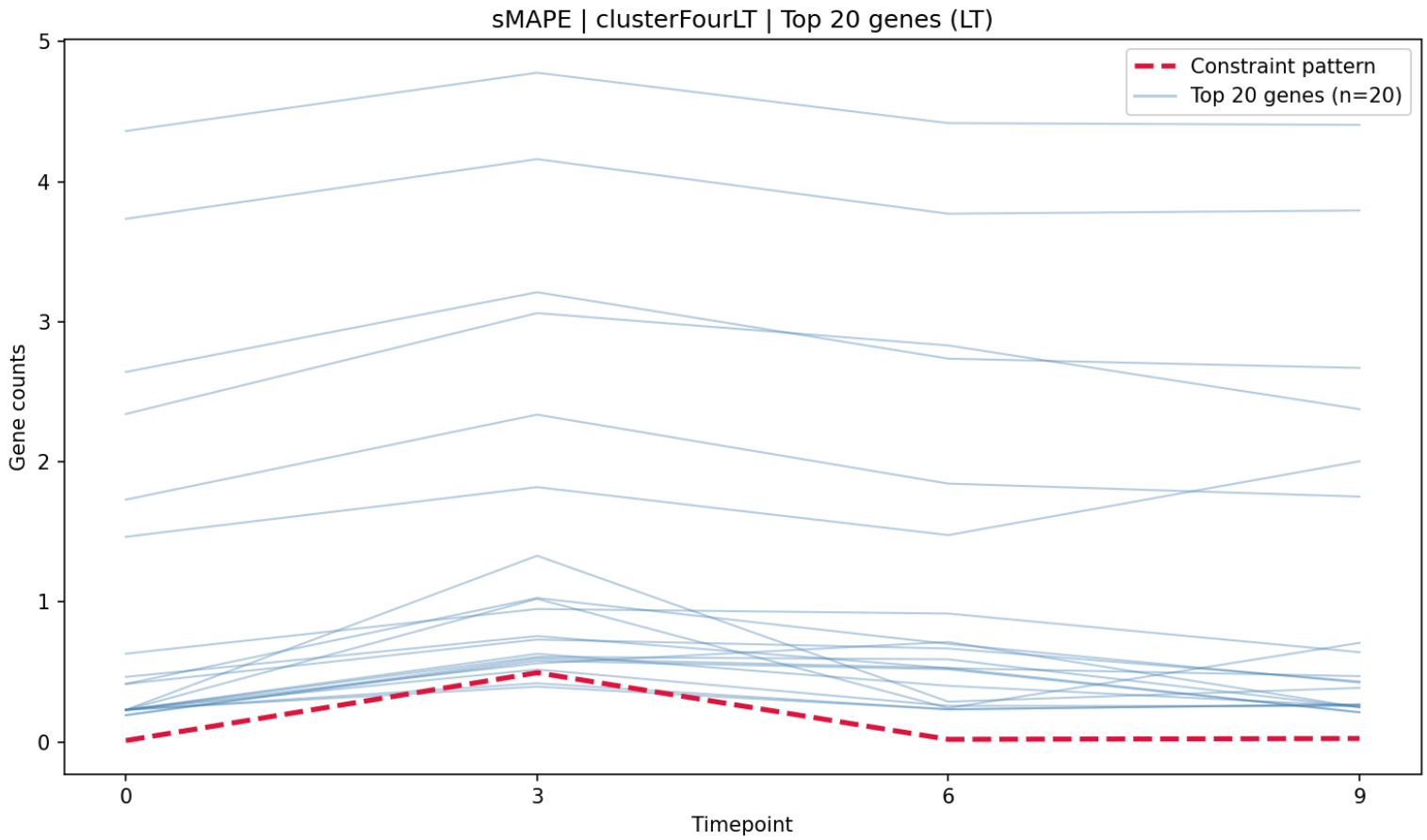
clusterFour LT — Fréchet



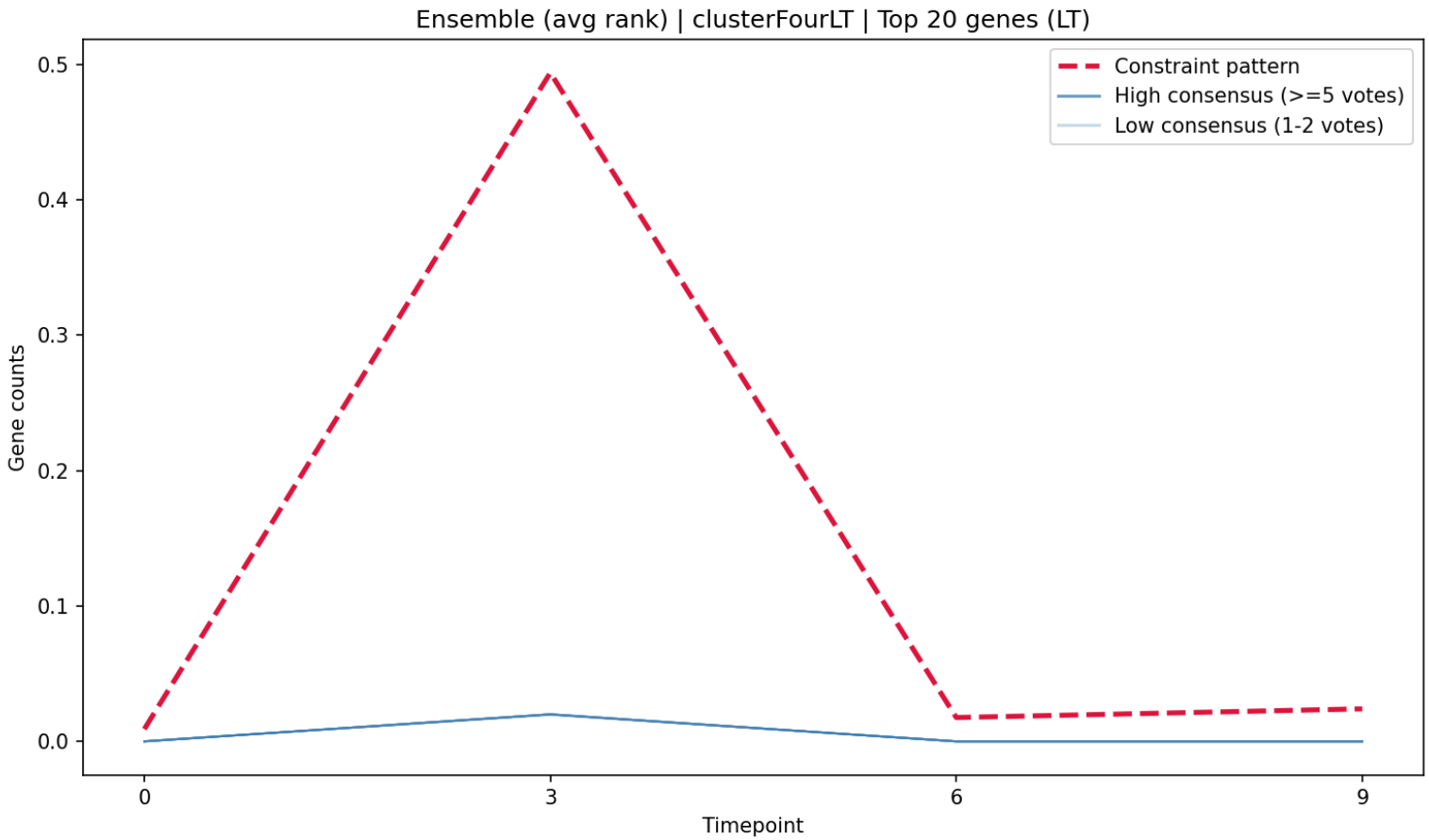
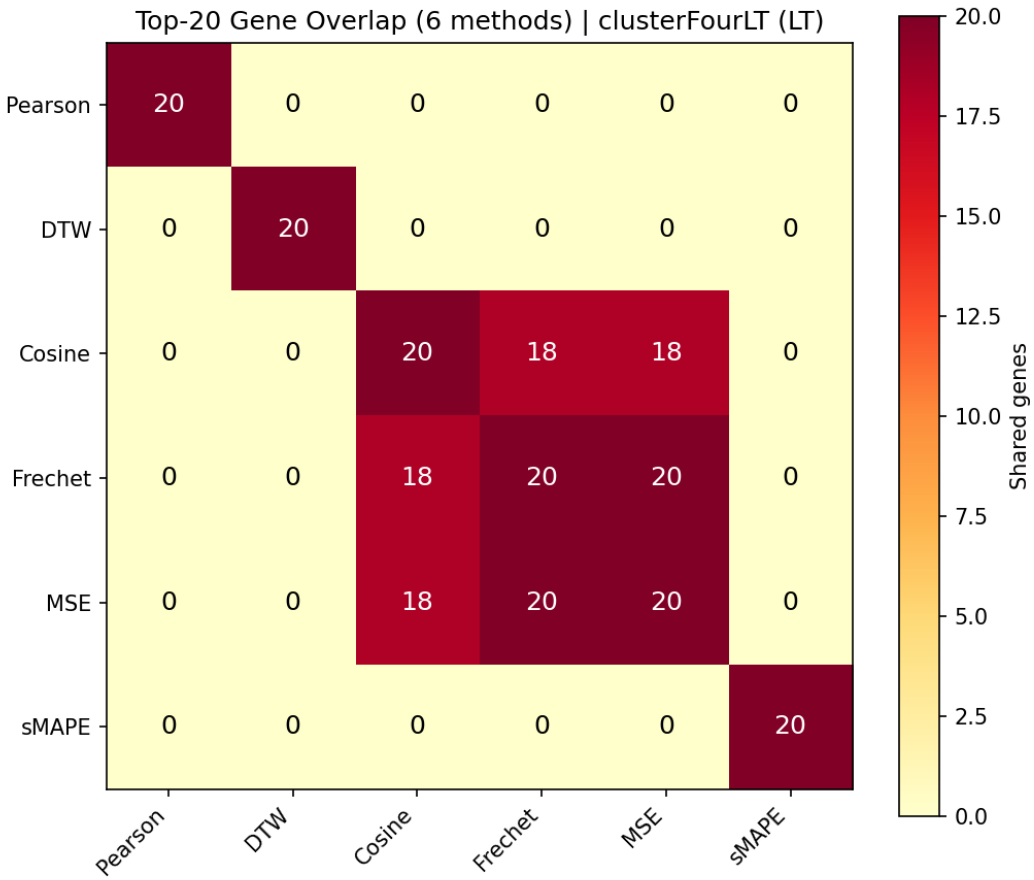
clusterFour LT — MSE



clusterFour LT — sMAPE



clusterFour LT — Ensemble



clusterFour LT — Overlap Heatmap